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PLATING POSTERS INC · METAL FINISHING REFERENCE SERIES

DECORATIVE CHROME

THE COMPLETE TRAINING MANUAL

A full training course for the brand-new operator — from "what is an anode?" to a signed certificate of completion. The bright, mirror chrome finish on nickel — covering both traditional hexavalent and modern trivalent chrome. Assumes you know nothing. Teaches you everything — including why the nickel underneath is where the real work happens.

01 **CLEAN** ▸ 02 **RINSE** ▸ 03 **ACID ACT.** ▸ 04 **RINSE** ▸ 05 **NICKEL**

06 **RINSE** ▸ 07 **CR ACT.** ▸ 08 **CHROME** ▸ 09 **RINSE** ▸ 10 **DRY/INSP.**

OPERATOR TRAINING & CERTIFICATION

EDITION 1.0 · 2026 · BEGINNER-FRIENDLY STYLE

Training reference only. If your shop runs **hexavalent decorative chrome (Cr VI), that bath is a confirmed human carcinogen** — treat it exactly as seriously as a hard-chrome tank. Always verify exact parameters against your chemistry supplier's Technical Data Sheets (TDS), customer specifications, and current Safety Data Sheets (SDS), and comply with OSHA 29 CFR 1910.1026 (Cr VI), EPA, REACH, and local regulations before production use.

FIND YOUR WAY

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★ **REMEMBER**

This manual is a training and reference tool — **not** a substitute for your shop’s written procedures, your supplier’s TDS/SDS, your OSHA Cr(VI) compliance program (if you run hexavalent), or your supervisor. When this manual and your shop’s official procedure disagree, **follow your shop’s procedure and ask your supervisor.**

PART ONE — FRONT MATTER & FUNDAMENTALS

START HERE

WELCOME ABOARD

IF YOU DON'T KNOW AN ANODE FROM AN APRON, YOU'RE IN EXACTLY THE RIGHT PLACE — AND HERE YOU'LL LEARN WHY A BRIGHT "CHROME" PART IS MOSTLY NICKEL

Welcome. If you're holding this manual, you're about to learn how to run — or at least understand — a **decorative chrome plating line**. Maybe you started this week. Maybe you've been racking parts for a year and finally want to know *why* you do the things you do. Either way, you're in the right place, and we're glad you're here.

Here's the most important thing we can tell you up front, and it's the idea this whole book is built around: **the bright "chrome" finish everyone admires is mostly nickel.** The chromium layer on a chrome bumper, a faucet, or a motorcycle fender is *astonishingly* thin — typically only **0.13 to 0.5 micrometers** (we'll define that). The thick, hard-working layer underneath it is **nickel**, and the nickel is what actually protects the part from corrosion. The chrome on top is a thin, bright, tarnish-resistant skin. Learn that one fact and half of this manual will already make sense.

The second thing: **this manual assumes you know nothing about plating, chemistry, or electricity.** That's not an insult — it's a promise. Every term gets defined the first time we use it, and every step explains not just *what* to do but *why* it matters. It's written in the spirit of the *plain-language* guides — we'd rather over-explain than leave you guessing in front of a tank full of acid.

> ⚠ **HEADS-UP — TWO VERY DIFFERENT CHROME BATHS EXIST, AND ONE IS A CARCINOGEN**
>
> Decorative chrome can be plated from **two completely different chemistries: hexavalent (Cr VI)** and **trivalent (Cr III)**. They look similar on the part but are worlds apart in hazard. **If your shop runs hexavalent decorative chrome, that bath is a confirmed human carcinogen** — the *same* hazard as a hard-chrome tank, even though the layer is far thinner. This manual teaches both. **Find out which one your shop runs before your first shift, and treat a hexavalent tank with total respect.**

> ★ **REMEMBER**
>
> You don't need to memorize this manual. You need to *understand* it. Understanding sticks. Memorizing fades by Friday.

• HOW TO USE THIS MANUAL

The manual follows the **line itself**, station by station, in the order a part actually travels: clean it, rinse it, activate it, rinse it, plate it with **nickel**, rinse it, activate it again, flash it with **chrome**, rinse it, and dry/inspect it. Reading it in order matters — the order of a plating line is never random, and neither is the order of this book.

> 💡 **TIP**
>
> Read the Fundamentals (Part One) all the way through *before* any station chapter. The station chapters assume you already understand the big picture — especially the nickel-undercoat system and, if you run it, the Cr VI hazard — that Part One gives you.

THE ICONS — YOUR FRIENDLY MARGIN HELPERS

> 💡 **TIP**
>
> A shortcut or trick of the trade — the kind of thing a 20-year veteran would lean over and tell you.

> ★ **REMEMBER**
>
> A core idea worth locking in. If you forget everything else on the page, don't forget these — they're the load-bearing ideas.

> ⚠ **HEADS-UP**
>
> A safety warning. Read every single one — these keep your skin, eyes, and lungs intact. We use them *liberally*, and on a hexavalent line *very* liberally.

> ⚙ **TECHNICAL STUFF**
>
> Deeper chemistry for the curious. Optional — skip it and you can still run the line; read it and you'll understand it on a deeper level.

> ☀ **FUN FACT**
>
> A bit of trivia to make the science stick (and to give you something to say at lunch).

Two more things at every station. Each station chapter ends with a **Teach-Back** checklist (the things you must be able to say out loud, unprompted, before you're cleared) and a bordered **Sign-Off** box your *trainer initials* once you've demonstrated the skill safely.

WHAT YOU'LL BE ABLE TO DO

LEARNING OBJECTIVES

MASTER THIS AND EVERY STATION CHAPTER CLICKS INTO PLACE

By the time you finish this manual, you should be able to confidently:

1. **Explain what electroplating is** in plain language — how electricity moves metal onto a part.
2. **Describe what decorative chrome is and what it's for** — a thin, bright, tarnish- and corrosion-resistant finish over a nickel undercoat, used for appearance and protection on consumer goods.
3. **Explain the single most important idea on this line:** the **nickel undercoat does the corrosion protecting; the chromium is a thin bright cap.** Decorative "chrome" is really a *nickel-chromium system.*
4. **Name the stations of the line in order** and explain *why* the order can't be rearranged.
5. **Tell hexavalent (Cr VI) and trivalent (Cr III) decorative chrome apart** — chemistry, color, control, cost, and especially hazard — and state the central safety fact: **a hexavalent bath is a confirmed human carcinogen.**
6. **Understand the nickel bath** — Watts/bright/semi-bright chemistry, pH, temperature, current density, and the brighteners that make it shine and level.
7. **Explain duplex/triplex nickel and micro-discontinuous chrome** — and *why* they dramatically boost corrosion life.
8. **Recognize the hazards at each station** — caustic cleaner, acids, **nickel sensitization and aerosol, boric acid, and (if hexavalent) Cr VI mist** — and know the correct PPE and emergency response.
9. **Use the basic field tests** every operator relies on — the water-break test, conductivity, and visual appearance.
10. **Talk the language** with your lead, your chemist, and your chemistry supplier.

> ★ **REMEMBER**
> You're not expected to hit all of these on day one. Plating is a craft. But the *safety* objectives (#5 and #8) are not optional and not gradual — you must have those before you work the line at all.

• THE LINE AT A GLANCE

Feel the rhythm: **Clean** → **Rinse** → **Acid Activate** → **Rinse** → **NICKEL PLATE** → **Rinse** → **Chrome Activate** → **CHROME PLATE** → **Rinse** → **Dry/Inspect.** Notice there are *two* highlighted tanks — the nickel (the real work) and the chrome flash (the bright cap).

> ★ **REMEMBER**
> Notice **two** highlighted tanks, not one. On a zinc or hard-chrome line there's a single star tank. On a decorative chrome line there are **two**: the **nickel** (which does the heavy lifting — thickness, brightness, leveling, corrosion protection) and the **chrome flash** (which adds the final bright, hard, tarnish-resistant skin). Most of the quality of a "chrome" part is decided in the **nickel** tank.

> ☀ **FUN FACT**
> Hard chrome and *hexavalent* decorative chrome use almost the same bath chemistry — chromic acid and a catalyst. The difference is *how much* you plate. Decorative chrome on a motorcycle is often only 0.25 µm thin over bright nickel; functional hard chrome can be hundreds of times thicker. Same dragon, very different job.

CHAPTER 1

WHAT IS ELECTROPLATING?

THE TRUE BEGINNER'S PRIMER — SO FAR BACK WE DON'T EVEN ASSUME YOU KNOW WHAT PLATING IS

THE ONE-SENTENCE VERSION

Electroplating is using electricity to coat one metal with a thin layer of another metal. That's the whole idea. Now let's unpack it.

Imagine a brass faucet body, or a steel motorcycle fender. It's strong and shaped right, but on its own it would tarnish, rust, or look dull. You want to give it a tough, bright, mirror-like skin — first of **nickel**, then a thin cap of **chromium**. You can't paint that on and get a real metal mirror. Instead, we use electricity to grow tightly-bonded layers of metal directly onto the part, atom by atom. That's electroplating.

HOW ELECTRICITY MOVES METAL

Picture a tank of liquid called the **electrolyte** (or "the bath"). It has metal dissolved in it as tiny charged particles called **ions**. (An *ion* is just an atom carrying an electrical charge — dissolved nickel floats around as **Ni^{2+}**.) You hook two things to a power supply called a **rectifier** (it converts ordinary AC wall power into the steady one-direction DC current plating needs):

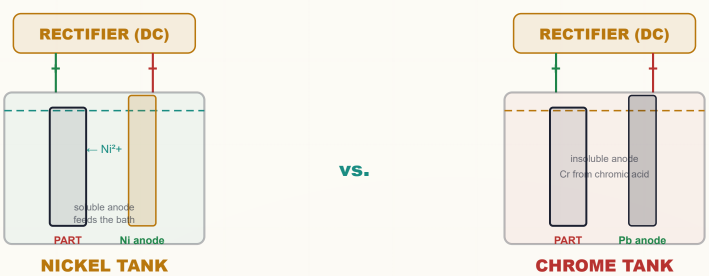
- **The cathode** — your *part*, connected to the **negative** terminal. *"Cathode = the part, the thing collecting metal."*
- **The anode** — bars or pieces hanging in the same tank, connected to the **positive** terminal.

When you turn on the rectifier, positively-charged metal ions are attracted to the negative part, reach the surface, grab electrons, and turn into solid metal stuck to the part. The longer you plate and the more current you push, the thicker the coating.

⚙ **TECHNICAL STUFF — SOLUBLE VS. INSOLUBLE ANODES, AND WHY THIS LINE HAS BOTH**

A decorative chrome line is unusual because it uses **both kinds of anode** at different tanks. In the **nickel** tank the anodes are **soluble nickel** — they dissolve to feed nickel back into the bath, just like zinc anodes feed a zinc bath. In the **hexavalent chrome** tank the anodes are **insoluble lead/lead-tin** — they carry current and keep the bath oxidized but do *not* dissolve; all the chromium comes from the chromic acid you maintain as a chemical. (Trivalent chrome uses yet another kind — usually special **inert/graphite or coated** anodes in a different cell design; more in Chapter 5.) Knowing *which anode feeds which tank* prevents a lot of confusion.

TWO TANKS, TWO KINDS OF ANODE (NOT TO SCALE)



★ **REMEMBER**

Three words per tank. **Nickel tank:** cathode = part, anode = *soluble nickel* (feeds metal), electrolyte = nickel salts. **Hex chrome tank:** cathode = part, anode = *insoluble lead* (carries current only), electrolyte = chromic acid (supplies the chromium). Almost everything on this line is a variation on these two ideas.

⚠ **HEADS-UP**

Plating rectifiers push enormous current at low voltage into tanks of hot, corrosive, conductive liquid. **Never reach into an energized tank, and never do maintenance until the rectifier is locked and tagged out (LOTO).**

THE DANCE, STEP BY STEP

When you energize the nickel tank, here's what physically happens:

1. Current flows. Positively-charged nickel ions (Ni^{2+}) in the bath are *attracted* to the negatively-charged part (the cathode), because opposite charges attract.
2. When a nickel ion reaches the part's surface, it grabs electrons and turns back into solid nickel metal, sticking to the surface.
3. Meanwhile, at the soluble nickel anodes, solid nickel dissolves *into* the bath, releasing fresh Ni^{2+} ions to keep the supply topped up.

The net result: nickel moves off the anodes, through the bath, and onto your part. The chrome tank works the same way for the cathode (chromium deposits on the part), but its anodes are *insoluble* — they don't feed the bath, so all the chromium has to come from chromic acid you maintain as a chemical.

⚙ **TECHNICAL STUFF**

At the part in the nickel tank, a **reduction**: $Ni^{2+} + 2e^- \rightarrow Ni^0$. At a soluble nickel anode, the reverse — solid nickel gives up two electrons and dissolves: $Ni^0 \rightarrow Ni^{2+} + 2e^-$. Because the anodes replenish what the cathode consumes, a well-run nickel bath is largely self-balancing on nickel metal. The chrome tank can't do this — its lead anodes don't dissolve — so its chromium is replenished by chemical additions, not by the anodes.

☀ **FUN FACT**

The unit "volt" traces back to Alessandro Volta, the 1790s Italian scientist who built the first true battery — the "voltaic pile" — out of stacked metal discs. You'll read volts on your rectifier every single day, named after a man who never saw a chrome bumper.

★ **REMEMBER**

Everything downstream is built on this one picture: **current pulls metal ions out of the bath and lays them on your part**. Get the part clean and well-connected, give the bath the right chemistry and the right current, and metal grows where you want it. Everything else in this manual is detail on top of that idea.

⚠ **HEADS-UP**

Both tanks are electrically "live" while plating, and both sit in conductive, corrosive liquid. Wet hands plus high-amperage bus bars is a serious shock hazard. Keep your hands dry, keep tools out of energized tanks, and never lean a metal rack across the bus bars.

CHAPTER 2

WHAT IS DECORATIVE CHROME?

THE FINISH EVERYONE CALLS “CHROME” IS MOSTLY NICKEL — AND THAT’S THE WHOLE POINT

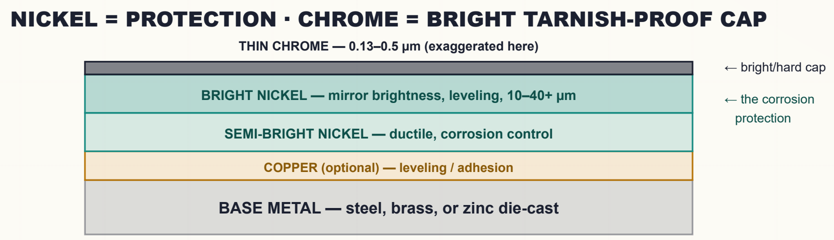
CHROME PLATING COMES IN TWO VERY DIFFERENT JOBS

“Chrome plating” can mean two completely different things:

- **Decorative chrome** — what *this* manual is about. An **extremely thin** bright flash of chromium (typically **0.13–0.5 µm**) over a base of **bright nickel**, used for shine, tarnish resistance, and corrosion protection on car trim and wheels, faucets and bath fixtures, motorcycle and bicycle parts, appliance handles, tools, and furniture. The **nickel underneath does almost all of the corrosion work**; the chrome is a thin, hard, decorative, tarnish-resistant cap.
- **Hard (functional/industrial) chrome** — the subject of our companion manual. A **thick, functional** chromium deposit plated **directly on steel** (no nickel underneath), valued for performance: extreme hardness, wear resistance, low friction, and rebuilding worn parts to size.

> ★ **REMEMBER**
> **Decorative chrome is a *system*, not a single layer.** It’s usually **base metal → (copper, sometimes) → nickel → thin chrome.** When someone says “chrome plating,” picture the *nickel-chromium system,* because that’s what’s really protecting and beautifying the part. The chrome you see is the thinnest layer in the stack.

THE LAYER STACK — WHAT’S REALLY ON A “CHROME” PART



LAYER	TYPICAL THICKNESS	WHAT IT DOES
Chromium (top)	`0.13–0.5 µm`	Bright, hard, tarnish/scratch-resistant cap; cosmetic + first corrosion barrier
Bright nickel	`10–20 µm (consumer); 20–40+ µm (auto)`	Mirror brightness, leveling, corrosion protection
Semi-bright nickel (duplex)	`part of 20–40 µm total`	Sulfur-free, ductile, sacrificial corrosion control (Part Three)
Copper (optional)	`varies`	Leveling, adhesion, cost; covers base-metal roughness
Base metal	–	Steel, brass, or zinc die-cast

> ★ **REMEMBER**
> Look at those numbers. The chrome is roughly **one-fortieth to one-hundredth** the thickness of the nickel. If you ever wonder where to focus your attention on this line, it’s the nickel. **A chrome part lives or dies on the nickel underneath it.**

WHAT DECORATIVE CHROME GIVES A PART

PROPERTY	WHAT IT MEANS ON THE FLOOR
Brightness / appearance	The mirror “chrome” look customers pay for
Tarnish resistance	Chromium does not yellow or tarnish like bare nickel does
Scratch / wear resistance	Chromium is hard (~800–1000 HV), so the bright finish survives handling
Corrosion resistance (system)	Mostly from the **nickel**; chrome adds a thin barrier and, when **micro-discontinuous**, dramatically improves it (Part Three)
Low maintenance	Easy to wipe clean; resists fingerprints and mild chemicals

Where you’ll see it: automotive trim, grilles, and wheels; faucets, showerheads, and bath hardware; motorcycle and bicycle components; appliance trim and handles; hand tools; lighting; furniture and hardware.

☀ **FUN FACT**

Bare nickel is actually a slightly *warm, yellowish* mirror. The thin chrome flash on top is what gives “chrome” its cool, bluish-white tone. Strip the chrome off a faucet and the nickel underneath still shines — it just looks a little yellow and will tarnish over time. The chrome is the tarnish-proof sunglasses on top of the nickel mirror.

WHY YOU CAN’T JUST PLATE THICK CHROME AND SKIP THE NICKEL

You might ask: if chromium is hard and bright, why not just plate it thick directly on the part and skip the expensive nickel? Two reasons:

1. **Thin chrome is porous and microcracked.** A decorative chrome layer is full of tiny pores and cracks — moisture finds its way straight through to the base metal, which then corrodes. The chrome alone is *not* a corrosion barrier.
2. **Chrome has terrible throwing power and is hard to build thick decoratively** while staying bright. Building thick chrome is the hard-chrome process — different bath control, different goal, and it isn’t bright.

So the division of labor is: **nickel provides thickness, leveling, brightness, and corrosion protection; chrome provides the final bright, hard, tarnish-resistant skin.** Each does what it’s best at.

⚙ **TECHNICAL STUFF — THE POROSITY PARADOX**

Here’s the elegant twist you’ll meet again in Part Three. Thin chrome’s porosity sounds like a flaw — and on its own it is. But **modern decorative chrome is deliberately engineered to spread that porosity into a fine, uniform pattern** (microcracked or microporous chrome). When the pores/cracks are *spread out evenly* over a good nickel undercoat, corrosion is forced to spread out too, instead of digging one deep pit. Spread-out, shallow attack looks far better and lasts far longer than a few deep pits. Engineers turned a weakness into a feature.

★ **REMEMBER**

The whole reason this line has so many stations is to lay down a *flawless nickel deposit* and then cap it with a *perfect bright chrome flash.* If you understand that goal, every clean, rinse, and dip on the line earns its place.

CHAPTER 3

HOW A CHROME LINE WORKS

THE BIG PICTURE — A LINE IS A SYSTEM, NOT A PILE OF SEPARATE TANKS

A decorative chrome line is a **sequence of stations** a part travels through in strict order. Each station does one job and prepares the surface for the next. Parts may be **rack plated** (hung individually — common for trim, wheels, and large parts) or **barrel plated** (tumbled in a rotating perforated drum — common for small hardware), depending on size and shape.

• THE WHOLE DECORATIVE CHROME SEQUENCE

#	STATION	THE ONE THING IT DOES	TEMP	NOTES
01	Clean / Degrease	Remove all oil, grease, shop soil	50–82°C	Alkaline soak ± electroclean
02	Rinse	Wash off cleaner (the firewall)	Ambient	—
03	Acid Activation	Remove oxide; wake up the surface	Ambient	Mild acid dip
04	Rinse	Guard rinse before nickel	Ambient	Protect the nickel bath
05	**Nickel Plate**	Build bright/duplex nickel undercoat	~55–65°C	**The real work; corrosion + brightness**
06	Rinse	Wash nickel film off before chrome	Ambient	Often a recovery/drag-out rinse
07	Chrome Activation / Strike	Activate nickel so chrome bonds & covers	in/near bath	Critical for coverage
08	**Chrome Plate**	Thin bright chrome flash (0.13–0.5 µm)	hex ~35–45°C / triv ~25–45°C	**Bright cap; hex = Cr VI carcinogen**
09	Drag-Out / Rinse	Recover dragged-out chrome; wash part	Ambient	Recovery matters (cost + hazard)
10	Dry / Final Inspection	Dry and verify appearance/thickness	–	Ship

⚠ **HEADS-UP — THE TEMPERATURES ARE DIFFERENT FROM HARD CHROME**

Decorative **hexavalent** chrome runs *cooler* than hard chrome — typically **~35–45°C** versus hard chrome's ~50–60°C. **Trivalent** chrome runs cooler still, often near room temperature to ~45°C. The **nickel** tank, by contrast, is genuinely hot (~55–65°C). Cooler hex chrome still throws **Cr VI mist** — cooler does not mean safe. Hot nickel throws nickel-bearing aerosol. Know which tank you're at.

RACK VS. BARREL

- **Rack plating:** each part hung on a current-carrying rack. Used for trim, wheels, fenders, faucets — larger, higher-value, or appearance-critical parts where position and contact matter.
- **Barrel plating:** many small parts tumbled in a rotating perforated drum. Used for fasteners and small hardware. Less common for show-finish chrome because tumbling can mar a mirror surface, but used for bright small hardware.

★ **REMEMBER**

Every station either protects the next or poisons it. There is no neutral step. A part that leaves the cleaner still oily contaminates the activation and won't take nickel. Nickel film dragged into the chrome tank can ruin coverage. Chrome dragged out untreated is wasted money — and, if hexavalent, a carcinogen loose in your shop. What you do (or fail to do) early shows up as a defect — or a safety failure — later.

💡 **TIP**

Picture the line as two acts. **Act One** (Stations 01–06) is all about building a *flawless nickel deposit.* **Act Two** (Stations 07–10) is about putting a *perfect bright chrome cap* on it. If Act One is bad, no amount of chrome can save the part. Get the nickel right and the chrome is almost easy.

CHAPTER 4

WHY THE ORDER MATTERS

WALKING THE LINE — EACH STATION SETS UP THE NEXT, AND THE SEQUENCE IS LOCKED BY CHEMISTRY

CLEAN BEFORE EVERYTHING.

Nickel can only bond to clean, bare metal. Plate over oil and you've plated over oil — the nickel bonds to grease and blisters or peels. The free, foolproof **water-break test** (clean metal sheets water in an unbroken film; dirty metal beads it) tells you when cleaning is done.

RINSE BETWEEN CLEANER AND ACID.

The cleaner is alkaline; the activation is acidic. Carry one into the other and you neutralize and waste both.

ACID-ACTIVATE RIGHT BEFORE NICKEL.

Even clean metal grows an invisible oxide film in minutes. A mild acid dip removes that oxide and "wakes up" the surface so nickel bonds. Freshly activated metal re-oxidizes fast — so activation comes immediately before the nickel tank.

NICKEL BEFORE CHROME — ALWAYS.

This is the heart of decorative chrome. The nickel is the corrosion barrier, the brightness, and the leveling. Chrome is a thin cap *on* the nickel.

RE-ACTIVATE THE NICKEL BEFORE CHROME.

Here's a subtle, critical point new operators miss: **nickel passivates (forms a thin invisible oxide) very quickly after it leaves the nickel tank.** Chrome won't bond or cover well on passivated nickel — you get bare spots, "skip plate," or poor adhesion. So before chroming, the nickel is **re-activated** (Station 07) — either by a short acid/cathodic strike or by the leading edge of the chrome bath itself. **Time matters: nickel that sits too long before chrome plates poorly.**

★ **REMEMBER**

No activation, no adhesion — and on this line you activate *twice*: once before nickel (Station 03) and again before chrome (Station 07), because **nickel passivates fast.** A part that sits between the nickel and chrome tanks is a part headed for a chrome coverage defect.

Plate chrome, then rinse, then dry/inspect — rinse to recover dragged-out chrome and stop it spreading (especially if hexavalent), then a *gentle* dry that won't water-spot the mirror finish, then inspect.

⚙ **TECHNICAL STUFF — THE RECURRING VILLAIN: DRAG-OUT**

Drag-out is the film of solution clinging to a part as it leaves a tank; **drag-in** is that film contaminating the next tank. Drag-out justifies almost every rinse on the line. On a *hexavalent* chrome line it's doubly serious: it wastes costly chromic acid *and* spreads a carcinogen. And **nickel drag-in to the chrome tank** is a specific enemy — nickel contamination is bad for chrome coverage. That's why the rinses around the nickel and chrome tanks are not optional.

⚠ **HEADS-UP**

Don't let a nickel-plated part sit wet and idle waiting for the chrome tank. Besides passivating (a quality problem), it can begin to spot or stain, and on a hexavalent line you want parts moving deliberately, not lingering. Smooth, prompt movement is both a quality habit and a safety habit.

CHAPTER 5

HEXAVALENT VS. TRIVALENT

TWO WAYS TO PUT A BRIGHT CHROME CAP ON NICKEL — AND ONE OF THEM IS A CARCINOGEN

The chrome flash (Station 08) can come from one of two completely different chemistries. They produce a similar-looking bright finish but differ in hazard, color, control, cost, and equipment. **You must know which one your shop runs.**

HEXAVALENT DECORATIVE CHROME (CR VI)

The traditional bath, used for over a century. It's **chromic acid (chromium trioxide, CrO_3) dissolved in water** plus a small amount of **catalyst** (sulfate, or a mixed catalyst). When CrO_3 dissolves, the chromium is in its **+6 ("hexavalent") oxidation state** — and **hexavalent chromium is a confirmed human carcinogen.**

- **Color:** a slightly **bluish-white**, "classic chrome" tone many consider the benchmark for appearance.
- **Coverage / throwing power:** moderate; decades of process know-how exist.
- **Hazard: the headline.** Cr VI mist is carcinogenic; the same OSHA Cr VI rules that govern hard chrome apply.

> ⚠ **HEADS-UP — HEXAVALENT DECORATIVE CHROME IS JUST AS CARCINOGENIC AS HARD CHROME**
>
> Do not let the *thin* deposit fool you. The **bath** is the hazard, not the coating thickness. A decorative hex tank is **chromic acid that throws an inhalable, carcinogenic Cr VI mist** — exactly like a hard-chrome tank, just run a bit cooler. **OSHA 29 CFR 1910.1026 applies**, with a **PEL of 5 µg/m³** and an **action level of 2.5 µg/m³**. Mist suppression, ventilation, respiratory protection, hygiene, and $Cr^{6+} \rightarrow Cr^{3+}$ waste treatment are all mandatory. If your shop runs hex decorative chrome, **read the Cr VI sections of this manual as carefully as a hard-chrome operator would.**

> ⚙ **TECHNICAL STUFF — "HEXAVALENT" VS. "TRIVALENT" IN ONE LINE**
>
> "Hexavalent" (Cr^{6+}, Cr VI) and "trivalent" (Cr^{3+}, Cr III) describe the chromium atom's oxidation state — how many electrons it's missing. **Cr(VI)** is a strong oxidizer, highly toxic, and carcinogenic. **Cr(III)** — the form chromium ends up in *after* being used or reduced — is far less hazardous and is even a trace nutrient. The entire goal of Cr VI waste treatment (Part Three) is to convert dangerous Cr^{6+} into safe Cr^{3+} and remove it.

TRIVALENT DECORATIVE CHROME (CR III)

A newer chemistry, increasingly common, built around **trivalent chromium salts** (chromium sulfate or chloride) plus complexing agents, conductivity salts, and buffers — *not* chromic acid. Because the chromium is already in the +3 state, **there is no carcinogenic Cr VI in the working bath.**

- **Color:** historically a slightly **warmer, darker, or "smokier"** tone than hex — though modern trivalent processes have closed much of that gap, and some are tuned to mimic the hex bluish-white. Color matching across a part run can be a control challenge.
- **Throwing power / coverage:** generally **better and more forgiving** than hexavalent — it covers low-current-density (recessed) areas more easily, which means fewer "skip plate" bare spots in recesses.
- **Hazard: much lower** — no Cr VI carcinogen in the bath — *but not zero.* Trivalent baths still contain **metal salts, boric acid, and complexers** that require proper handling, skin/eye protection, and waste treatment. And trivalent baths are **more sensitive to metallic contamination** (even small amounts of nickel, copper, iron, or zinc dragged in can poison them) and need tighter chemistry control.
- **Equipment:** different — usually **special inert/coated or graphite anodes** in a divided or shielded cell to prevent the bath oxidizing its own Cr^{3+} back up to Cr^{6+}.

> ★ **REMEMBER — THE TRIVALENT TRADE-OFF IN ONE SENTENCE**
>
> Trivalent chrome trades **lower hazard and better coverage** for **tighter chemistry control, contamination sensitivity, and a sometimes harder-to-match color.** It is *lower hazard, not no hazard* — boric acid, metal salts, and complexers still demand respect.

☀ **FUN FACT**

One way to tell a trivalent line from a hexavalent one without a lab: trivalent baths are often **darker green/blue and run cooler**, sometimes near room temperature, while a classic hexavalent bath is the familiar **orange-yellow** of chromic acid. Color is a clue — never a substitute for knowing your shop's documented process.

SIDE-BY-SIDE

FACTOR	HEXAVALENT (CR VI)	TRIVALENT (CR III)
Main chemistry	Chromic acid (CrO_3) + catalyst	Cr^{3+} salts + complexers + buffers
Carcinogen in bath?	**Yes — Cr VI, confirmed human carcinogen**	**No Cr VI in the working bath**
Color	Bluish-white "classic chrome"	Warmer/darker historically; modern processes closer to hex
Throwing power / coverage	Moderate	Generally **better / more forgiving**
Temperature	~35–45°C	~25–45°C (often cooler)
Anodes	Insoluble lead/lead-tin	Special inert/coated or graphite
Contamination sensitivity	Tolerant	**Sensitive** (metals poison it)
Regulatory burden	High (OSHA 1910.1026, REACH)	Much lower
Maturity / color consistency	Mature, well-understood	Newer; color control needs attention

⚠ **HEADS-UP**

Knowing that a *trivalent* alternative exists does **not** change how you must treat a *hexavalent* bath in front of you. If the tank you run is Cr(VI), treat it as a carcinogen, every shift — exactly as a hard-chrome operator does.

💡 **TIP**

This choice — hex or trivalent — is a *process-engineering decision* made by your shop, not by you at the tank. Your job is to know which one you run, treat its specific hazards correctly, and control its specific quirks (Cr VI mist for hex; contamination and color drift for trivalent). Part Three returns to the decision in more depth for supervisors.

★ **REMEMBER**

If you take one thing from this chapter: **find out, before your first shift, whether your chrome tank is hexavalent or trivalent.** Everything about how you protect yourself at Station 08 flows from that single answer.

CHAPTER 6

THE ELECTROCHEMISTRY

TWO VERY DIFFERENT TANKS — A WORKHORSE NICKEL BATH AND A STRANGE CHROME BATH

This line has two plating tanks that behave nothing alike. Understanding both makes everything downstream make sense.

• THE NICKEL TANK — A WELL-BEHAVED WORKHORSE

The nickel bath is, by plating standards, *friendly.* It's efficient, fast, and forgiving compared to chrome.

- **Cathode efficiency: ~93–97%** — nearly all your current makes nickel, very little waste. (Compare chrome, below.)
- **Anodes: soluble nickel** — they dissolve to feed nickel back into the bath, so the bath is largely self-feeding for metal.
- **Throwing power: good** — nickel covers recesses and far surfaces reasonably well, which is one more reason the nickel (not the chrome) carries the corrosion protection.
- **Deposit:** bright, leveled, ductile, and built to useful thickness (10–40+ µm).

TECHNICAL STUFF — WATTS NICKEL, THE CLASSIC RECIPE

The most common bright-nickel base is a **Watts nickel** bath: **nickel sulfate** (the main nickel source), **nickel chloride** (helps the anodes dissolve and boosts conductivity), and **boric acid** (a *buffer* that holds the pH steady right at the part surface so the deposit stays bright and sound). On top of that go **brighteners and levelers** (organic additives) and **wetting agents**. Typical ranges: nickel sulfate ~225–300 g/L, nickel chloride ~40–60 g/L, boric acid ~30–45 g/L, **pH ~3.8–4.5**, **temperature ~55–65°C**, **current density ~3–5 A/dm²**. Your exact numbers come from your supplier's TDS.

TECHNICAL STUFF — WHAT BRIGHTENERS AND LEVELERS ACTUALLY DO

Brighteners are organic additives that make the deposit reflective by refining the grain (Class I "carriers" and Class II "brighteners" work together). **Levelers** preferentially build metal into tiny scratches and valleys, so the surface comes out *smoother than it went in* — that's how a slightly dull buffed part becomes a mirror. This is why a great chrome finish is mostly a great **bright nickel** finish: the leveling and brightness happen in the nickel tank, not the chrome tank.

• THE CHROME TANK — ELECTROCHEMICALLY STRANGE

Hexavalent chrome behaves much like the hard-chrome bath, just thinner and cooler:

- **Cathode efficiency: low — roughly 10–20%.** Most of your current splits water into **hydrogen** at the part and **oxygen** at the anode, not chrome. (This is why the chrome flash is thin and why it makes gas.)
- **Current density: high** — typically **~10–25+ A/dm²** for decorative work (lower than hard chrome but still much higher than nickel).
- **Anodes (hex): insoluble lead/lead-tin** — carry current, keep the bath oxidized; the chromium comes from the chromic acid as a chemical.
- **Throwing power: poor** — chrome piles onto high spots and starves recesses, giving the classic coverage challenges of Station 08.
- **Catalyst ratio (hex):** the bath needs a small, controlled amount of **catalyst (sulfate)**, classically **CrO_3 : SO_4 ≈ 100:1** by weight — the same "heart of the bath" ratio as hard chrome.

Trivalent chrome is different again: lower current density, near room temperature, special anodes, and the better throwing power noted in Chapter 5 — but it demands tight chemistry and contamination control.

★ REMEMBER

Two tanks, two personalities. Nickel: high efficiency (~93–97%), soluble anodes, good throwing power, does the real work. Hex chrome: low efficiency (~10–20%), insoluble lead anodes, poor throwing power, thin bright cap, **and a carcinogen.** If you carry those two contrasts in your head, you'll understand most of what happens on this line.

⚠ **HEADS-UP — HYDROGEN AT THE CHROME (AND NICKEL) TANK**

The chrome bath's low efficiency makes a lot of **hydrogen** at the part — hydrogen is flammable and, on high-strength steel, can cause **hydrogen embrittlement** (the part becomes brittle and can crack under load). Decorative parts are usually mild steel, brass, or zinc die-cast where embrittlement is less of a concern, but **on high-strength steel a post-plate bake may be required** — follow the print and your shop's spec. (More in Part Three.)

KEY TERMS TO CARRY FORWARD

- **Cathode efficiency:** the percent of current that becomes metal. Nickel ~93–97%; chrome ~10–20%.
- **Throwing power:** the bath's ability to plate evenly into recesses. Nickel good; chrome poor.
- **Leveling:** building metal preferentially into valleys so the surface comes out smoother (a nickel-tank job).
- **Duplex/triplex nickel:** two or three nickel layers with different chemistries for corrosion control (Part Three).
- **Micro-discontinuous chrome:** chrome with a controlled fine pattern of cracks (microcracked) or pores (microporous) to spread corrosion out (Part Three).
- **Boric acid:** the pH buffer in the nickel bath — and a handling hazard (Chapter 7).

☀ **FUN FACT**

The nickel tank's ~95% efficiency means almost every amp-hour you push becomes nickel on the part. The chrome tank's ~15% means most of your amp-hours go into making bubbles. The same rectifier, the same operator — but one tank is a generous workhorse and the other is a spendthrift that mostly makes gas. That's why the chrome layer is so thin: it's expensive, slow current-wise, and only needs to be a skin.

⚙ **TECHNICAL STUFF — WHY BORIC ACID MATTERS IN THE NICKEL BATH**

At the part surface, plating consumes hydrogen ions, so the local pH wants to *rise*. If it rises too far you get cloudy, rough, or burnt nickel and even nickel-hydroxide inclusions. **Boric acid buffers** the surface pH, holding it steady so the deposit stays bright and sound. That's why boric acid isn't optional and why bath pH is one of the most-checked numbers on the line — you'll meet this again at Station 05.

★ **REMEMBER**

When a "chrome" part comes out hazy, dull, or milky, your instinct may be to blame the chrome tank. Nine times out of ten the brightness and leveling problems live in the **nickel** tank. The chrome tank's job is coverage and a bright cap — the *mirror* is built in nickel.

CHAPTER 7

PPE — YOUR DAILY ARMOR

THE GEAR BETWEEN YOU AND HOT CAUSTIC, ACIDS, NICKEL, BORIC ACID, AND — IF HEXAVALENT — A CARCINOGENIC MIST

A decorative chrome line works with hot caustic cleaners, acids, **nickel salts**, **boric acid**, high-amperage current, and — if you run hexavalent chrome — **carcinogenic Cr VI chromic acid and its mist.** PPE is your *last* layer of protection (after ventilation and mist suppression), but it's essential.

★ **REMEMBER**

There is no part, no order, and no deadline worth your health. Scrap can be reworked. Your lungs and skin cannot.

THE MASTER PPE SET (WEAR AT ALL CHEMICAL STATIONS)

- **Chemical splash goggles** — sealed goggles, not safety glasses. Glasses leave gaps; acids and chromic acid find them.
- **Face shield** — *over* the goggles for work at the chrome tank, acid, or cleaner, and when charging chemistry.
- **Chemical-resistant gloves** — elbow-length, rated for the chemistry (nitrile, neoprene, or PVC per the SDS). Inspect for pinholes before each use.
- **Chemical-resistant apron/suit** — chest to below the knee.
- **Chemical-resistant boots** — closed, non-slip, splash-proof.
- **Respiratory protection** — required at a **hexavalent** chrome tank per your shop's Cr VI program; also used per your program for nickel aerosol where needed. Used *in addition to* (never instead of) ventilation.
- **Dedicated work clothing** — kept separate from street clothes; especially important on a hexavalent line, to keep Cr VI out of your car and home.

⚠ **HEADS-UP — NICKEL IS A SENSITIZER**

Nickel salts and nickel aerosol/mist can cause **allergic skin sensitization (nickel dermatitis)** and respiratory irritation; some nickel compounds are also classed as carcinogenic by inhalation. Once you're sensitized to nickel, you can react to it for life. **Keep nickel solution off your skin, wear your gloves, and use ventilation over the hot nickel tank** to avoid breathing nickel-bearing aerosol.

⚠ **HEADS-UP — BORIC ACID HANDLING**

The boric acid in the nickel bath is a buffer, but as a chemical it's a **reproductive-hazard concern** and a skin/eye/respiratory irritant — REACH flags it as a substance of very high concern. Don't handle boric acid powder without gloves and dust protection, avoid creating dust, and follow the SDS when making bath additions.

💡 **TIP**

Put gear on *in order:* boots and apron first, then gloves, then goggles, then face shield and respirator last. Take it off in reverse, and rinse your gloves *before* removing them. This keeps contaminated gloves away from your face — critical near nickel and Cr VI.

⚠ **HEADS-UP — RESPIRATOR IS A BACKUP, NOT A LICENSE (HEXAVALENT LINES)**

A respirator is required at a hex chrome tank, but it does **not** replace tank ventilation and mist suppression. If the fume suppressant has died (foam gone, mist visible) or the exhaust isn't pulling, the tank is unsafe even with a respirator on. Stop and report it. Engineering controls are primary; the respirator is the last line.

STATION-BY-STATION HAZARD SUMMARY

STATION	PRIMARY HAZARD	WHY IT'S DANGEROUS
01 Clean/Degrease	**Hot alkaline (caustic)**	Chemical burns; hot splash
02 Rinse	Residual cleaner	Mild irritant; slip hazard
03 Acid Activation	**Acid**	Burns; eye damage
04 Rinse	Residual acid	Mild irritant; slip
05 Nickel Plate	**Hot nickel bath; nickel sensitization/aerosol; boric acid**	Allergic dermatitis; respiratory; burns
06 Rinse	Nickel-bearing water	Sensitizer; slip
07 Chrome Activation	Acid/strike (or hex chrome)	Burns; possible Cr VI
08 Chrome Plate	**Hex: Cr VI mist + chromic acid (carcinogen).** Triv: metal salts, boric, complexers	**Hex: carcinogen, severe burns, respiratory. Triv: lower but real**
09 Drag-Out/Rinse	Chrome-contaminated water (Cr VI if hex)	Carcinogen (hex); burns; slip
10 Dry/Inspect	Residual chemistry; hot air	Mild burns; slip

⚠ **HEADS-UP**

Never eat, drink, smoke, vape, or apply cosmetics on the line. Hand-to-mouth contact is a major route for ingesting nickel and — on a hex line — Cr VI. Wash thoroughly before breaks and before leaving. Keep food and drink completely out of the work area.

💡 **TIP**

The two "quiet" hazards on this line are the ones new operators forget: **nickel sensitization** (a lifelong allergy you can develop from sloppy glove habits) and **boric acid** dust on additions. They don't burn or sting in the moment, so they're easy to ignore — which is exactly how people get hurt by them over time. Glove up and ventilate even when the tank seems harmless.

CHAPTER 8

EMERGENCY & FIRST-RESPONSE

WHEN SECONDS COUNT — LEARN THIS COLD, BEFORE YOU EVER NEED IT

Know these *before* you need them. **Locate, before your first shift:** the nearest **emergency eyewash** (within ~10 seconds of every chemical station), the **safety shower**, the **fire extinguisher** and alarm, the **rectifier emergency stop / disconnect**, the **spill kit**, the **SDS binder**, and — on a hexavalent line — your shop's **Cr VI regulated-area** boundaries and decon station.

> **HEADS-UP**
> **Eyewash and shower stations must always be unblocked.** Never stack parts, totes, or anode boxes in front of them. The day you need one is the day you can't move boxes out of the way.

• FIRST RESPONSE BY HAZARD

SITUATION	WHAT TO DO — IMMEDIATELY
Acid / caustic / nickel / chromic acid on skin	Safety shower; flush **at least 15 minutes.** Remove contaminated clothing while flushing. Even small chromic-acid contact on broken skin needs medical attention (chrome-ulcer risk). Bring the SDS.
Chemical in eyes	Eyewash; hold eyelids open and flush **at least 15 minutes,** rolling the eyes. Do not rub. Get medical attention immediately. Bring the SDS.
Inhaled chromic acid mist (hex), feel ill	Move to fresh air immediately. **Report it** — don't "tough it out." Seek medical attention for anything beyond brief mild irritation; report it as a possible Cr VI exposure so monitoring/medical follow-up happens.
Nickel rash / breathing irritation	Wash exposed skin; report developing rashes (possible nickel sensitization). Report respiratory irritation — it can signal inadequate ventilation.
Nosebleed / sore nasal passages (hex lines)	Report it — recurrent nosebleeds or nasal soreness can be an early sign of Cr VI over-exposure; it triggers a ventilation/exposure check.
Cr(VI) swallowed (hex)	**Do not** induce vomiting unless the SDS directs it. Follow SDS first aid; call poison control / medical help immediately.
Electrical shock / arc-flash	**Do not touch the person** if they may still be in contact with current. Cut power at the emergency stop/disconnect first, then call for help and render aid.
Chemical spill	Alert others; keep people away. Clean only if trained and within your shop's "small spill" limits. **Never let chromic-acid or nickel spills reach a floor drain** — that's a reportable environmental release.

> **HEADS-UP — LOCKOUT/TAGOUT (LOTO)**
> Mandatory for *all* rectifier and electrical maintenance. Physically lock the power off and tag it so no one can re-energize while someone works. "I'll just be a second" is how people get electrocuted on a high-amperage chrome or nickel rectifier.

> **HEADS-UP — A CRITICAL ACID-MIXING RULE, FOR LIFE**
> When diluting acid (or dissolving chromic acid per your procedure), **always add ACID/chemical to WATER — never water to concentrated acid.** Adding water to concentrated acid releases heat so violently it can erupt acid into your face. The memory aid: *"Do as you oughta — add acid to water."*

> **FUN FACT**
> The reason acid-to-water matters is heat capacity: a big pool of water soaks up the heat of dilution; a little water dropped onto acid flash-boils and spits. This one rule has saved more faces than any other in the chemical trades.

CHAPTER 9

THE SAFETY PHILOSOPHY

THE MINDSET THAT KEEPS YOU SAFE ACROSS A WHOLE CAREER ON A CHROME-OVER-NICKEL LINE

Rules and PPE are the *tools.* The philosophy is the *mindset* that makes them work.

1. KNOW WHICH CHROME YOU RUN — AND TREAT HEXAVALENT AS A CARCINOGEN, ALWAYS.

The single biggest safety question on this line is *hex or trivalent?* If it's hex, the chromic-acid mist is an *invisible* carcinogenic hazard; trust the controls (ventilation, foam blanket, air monitoring), not your senses.

2. ENGINEERING CONTROLS COME BEFORE PPE.

Suppress the mist at the source, ventilate the tank, *then* wear your respirator. PPE is the last layer, not the first.

3. RESPECT THE NICKEL, TOO.

It's easy to fixate on chrome and forget nickel is a **sensitizer** that can give you a lifelong allergy, and that boric acid is a real handling hazard. The "quiet" tank still demands gloves, ventilation, and hygiene.

4. HYGIENE IS A SAFETY CONTROL, NOT HOUSEKEEPING.

No eating, drinking, smoking, or face-touching; wash before breaks; keep work clothes separate. With both nickel and (if hex) Cr VI, hand-to-mouth and take-home contamination are real exposure routes.

5. CONTAINMENT BEATS CLEANUP.

Move parts deliberately; let them drain over the tank; don't fling racks or splash. Most exposures aren't dramatic spills — they're drips and aerosols from sloppy handling. Smooth movement is safer *and* makes better parts (less drag-out, less mist).

6. WHEN IN DOUBT, STOP AND ASK.

The unsafe operator isn't the one who asks questions — it's the one who guesses. If the foam looks thin, a glove looks worn, the exhaust sounds weak, or a reading looks off, *stop and check.*

• THE HAZARD FAMILIES TO INTERNALIZE

1. **Cr(VI) / chromic acid (if you run hexavalent — the headline).** Carcinogenic mist, corrosive liquid, chrome ulcers, nasal damage. *Defense:* mist suppression, ventilation, respirator, full PPE, hygiene, OSHA 1910.1026 program, $Cr^{6+} \rightarrow Cr^{3+}$ waste treatment.
2. **Nickel & boric acid.** Sensitization/dermatitis, nickel aerosol, boric-acid reproductive concern. *Defense:* gloves, ventilation, hygiene, dust control on additions.
3. **Acids & alkali (corrosives).** Hot caustic cleaner and acid dips burn on contact and damage eyes. *Defense:* goggles, face shield, gloves, apron; 15-minute flush; acid-to-water.
4. **Electrical & physical.** High-amperage rectifiers (shock/arc-flash); wet/slick floors; hot tanks. *Defense:* LOTO, dry hands, non-slip boots, heat awareness.

★ **REMEMBER**

You now have the whole foundation: what plating is, why decorative chrome is really a **nickel-chromium system**, how the line works, the two chrome chemistries (one a carcinogen), the two tank personalities, what to wear, what to do in an emergency, and the safety mindset that ties it together. Every station chapter builds on this. Turn the page knowing the *why* — and especially the *danger* — and the *how* will make a lot more sense.

PART TWO — STATION BY STATION

STATION 01 OF 10

CLEAN / DEGREASE

"PLATE OVER DIRT AND YOU'VE JUST PLATED DIRT."

★ **REMEMBER — THE PREP-LINE PROMISE**

By the time a part reaches the nickel tank, most of whether it'll turn out beautiful or scrap has already been decided. The prep stations — Clean, Rinse, Activate, Rinse — are not the "boring stuff before the real plating." As the old-timers say: *"The plating bath gets the credit, but the pretreatment does the work."*

Purpose. Get the part **chemically clean** — no oil, grease, fingerprints, polishing compound, or shop soil — so nickel bonds to bare metal.

• THE "WHY"

Nickel can only bond to clean, bare, active metal. Oil, buffing/polishing compound, and shop soil block the bond; nickel plated over grease will blister or peel, and the chrome on top fails with it. Decorative parts often arrive **buffed or polished**, leaving greasy polishing compound that must be removed — so cleaning here often means a **hot alkaline soak** plus **alkaline electrocleaning** (running current to scrub the surface electrochemically). The free, foolproof check is the **water-break test.**

💡 **TIP — THE WATER-BREAK TEST (FREE, FAST, NEVER LIES)**

Pull a cleaned part and watch the water. A truly clean surface sheets water in **one continuous, unbroken film** (PASS). A dirty surface makes water **bead up** (FAIL — re-clean). Costs nothing; takes two seconds; use it on every part.

```
FAIL - water beads up         PASS - water sheets evenly
+--------------+              +--------------+
|  o   o   o   |  X           |##############|  OK
|    o    o    |              |##############|
+--------------+              +--------------+
Oil remains -> RE-CLEAN       Surface clean -> proceed
```

⚙ **TECHNICAL STUFF**

"Water break" reflects the metal's *surface energy.* Clean metal has high surface energy so water spreads; even a one-molecule-thick oil film drops the energy so water beads. That's why it catches contamination far too thin to see.

⚙ **TECHNICAL STUFF — BUFFING COMPOUND IS THE DECORATIVE VILLAIN**

Decorative parts are frequently polished/buffed to a mirror *before* plating, and buffing wax/compound is a tenacious, greasy soil. Inadequate removal here is one of the most common causes of nickel blistering and peeling on show parts. When in doubt, clean longer and re-test.

• MAKE-UP & KEY PARAMETERS

VERIFY VS. YOUR TDS

PARAMETER	RANGE	WHAT TO WATCH
Type	Alkaline soak ± electroclean	Electroclean for heavy polishing compound
Temperature	50–82°C (120–180°F)	Per cleaner TDS; don't overheat
Time	2–10 min soak	Heavy buffing compound longer
Concentration	Per TDS (often 30–75 g/L)	Titrate regularly
pH	Strongly alkaline (~11–13.5)	Burns on contact
Electroclean current	Per TDS (anodic/cathodic/PR)	Polarity per part type
Water-break test	Pass = continuous sheet	The only reliable field test

Plain-language definitions: **Concentration** = how much cleaner chemical is dissolved per liter; too weak won't cut oil, too strong wastes money. **Titrate** = a simple lab test that measures actual concentration so you know when to add more. **Electroclean** = running current through the part in the cleaner so gas bubbles scrub the surface electrochemically — the strongest cleaning step for stubborn buffing compound.

⚠ **HEADS-UP — HOT ALKALINE SOLUTION**

This tank is **hot AND caustic.** Splashes cause **chemical burns on contact,** and steam/mist irritates airways. Full PPE: splash goggles, face shield (when charging/dumping), elbow-length chemical gloves, apron, boots. Skin contact → flush 15 min; eyes → eyewash 15 min and get help.

• STEP-BY-STEP PROCEDURE

1. **Check the tank:** temperature in range, concentration in range (last titration log), surface free of floating oil.
2. **Inspect the load** — heavy buffing compound or rust-preventive oil needs the long end of the time range.
3. **Immerse fully,** start the timer, confirm agitation.
4. **Electroclean if specified,** at the correct polarity for the part.
5. **Withdraw slowly,** draining back into the tank.
6. **Run the water-break test.** Continuous sheet → next station. Beading → re-clean.
7. **Move promptly** so the clean surface doesn't sit and flash-rust or re-soil.

• TOP MISTAKES

1. Skipping the water-break test because the part "looks shiny" (polishing compound is invisible once thin).
2. Running the tank too cold to cut buffing wax.
3. Letting concentration drift low.
4. Touching cleaned parts with bare or dirty gloves (re-contamination).
5. Letting clean parts sit and re-soil or flash-rust before activation.

• TEACH-BACK CHECKLIST

A trainee is signed off on Station 01 only when they can, unprompted:

- [ ] State what the cleaner removes and what it does *not* (oil/soil — not oxide; that's activation's job).
- [ ] Correctly perform and read a water-break test.
- [ ] State the temperature range and the caustic-burn hazard.
- [ ] Explain why buffing compound matters on decorative parts.
- [ ] Name three pieces of required PPE.

• QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Water break fails	Low concentration or temp; heavy buffing compound	Titrate cleaner; bring temp to range; add electroclean
Oily film after cleaning	Floating oil redepositing	Skim surface; dump tank if heavily contaminated
Blistering/peeling downstream	Buffing compound not fully removed	Longer clean + electroclean; re-test water break
Etching/discoloration	Cleaner too aggressive or too hot	Reduce temp; consider milder formulation

SIGN-OFF — STATION 01

Trainee: ______________________ Trainer: ______________________ Date: ____________

"I confirm the part passed the water-break test (continuous sheet), the cleaner was in range and at temperature, and I wore required PPE."

💡 **TIP**

On a decorative line, the cleaning station is where mirror finishes are won or lost. A part that fails the water-break here will fail the customer's eye later — as a blister, a haze, or a peel. Two seconds of testing now saves a stripped-and-replated part later.

STATION 02 OF 10

RINSE (THE FIREWALL)

"A RINSE ISN'T A REST STOP. IT'S A CHECKPOINT."

Purpose. Wash the **alkaline cleaner film** off the part before it reaches the acidic activation.

THE "WHY"

The cleaner is alkaline; the activation is acidic. Carry alkaline drag-out forward and it neutralizes and weakens the acid, wastes chemistry, and degrades activation — showing up later as poor adhesion. The rinse knocks that drag-out down by dilution. We measure rinse cleanliness with **conductivity (µS/cm)** — *lower = cleaner.*

TIP — COUNTER-FLOW RINSING

The professional approach is a **counter-current** rinse: parts move forward, water flows backward. Fresh water enters the *last* (cleanest) tank and overflows into the dirtier one, so the cleanest water touches the part last. It saves enormous water — important on a line that needs careful water management for nickel and (if hex) Cr VI control.

MAKE-UP & KEY PARAMETERS

PARAMETER	RANGE	NOTES
Temperature	Ambient	No heating needed
Time	20–60 s with agitation	Longer for recesses/blind holes
Water source	Municipal or DI	DI preferred
Conductivity	< 500 µS/cm (target < 300)	Monitor at shift start
Flow	Continuous overflow / counter-current	Prevents stagnation

HEADS-UP

The rinse carries dragged-in caustic and starts mildly alkaline. Wear goggles, gloves, non-slip footwear. **Wet floors and slips are the #1 everyday injury at rinses.**

STEP-BY-STEP PROCEDURE

1. Check conductivity at shift start; if near/above limit, tell your lead.
2. Confirm water is flowing.
3. Transfer the rack promptly from the cleaner (dried cleaner is harder to rinse).
4. Immerse fully with agitation, 20–60 s; longer for recesses.
5. Lift and drain over the rinse tank.
6. Move promptly to activation so the part doesn't flash-rust.

TEACH-BACK — CAN YOU ANSWER?

- Why does carrying cleaner into the acid cause problems?
- Define drag-out and drag-in.
- What does conductivity tell you — which way is "clean"?
- State the limit and the target number.
- Why rinse recesses longer?

TOP MISTAKES TO AVOID

- Treating the rinse as a "quick dunk."
- Ignoring the conductivity reading.
- Letting parts drip-dry between cleaner and rinse.
- Not flushing blind holes and recesses.
- Skipping the drain step.

SIGN-OFF — STATION 02

Trainee: ______________________ Trainer: ______________________ Conductivity: __________ µS/cm

"I confirm the rinse conductivity was within limit, water flow was running, and parts were rinsed the full time with PPE worn."

STATION 03 OF 10

ACID ACTIVATION

“WAKE THE METAL UP SO THE NICKEL WILL GRAB IT.”

Purpose. Remove the invisible oxide film and lightly activate the surface so nickel forms a true bond.

• THE “WHY”

Even spotlessly clean metal grows an oxide film in minutes, and oxide blocks adhesion. A **mild acid dip** (often dilute sulfuric or hydrochloric, or a proprietary acid salt) dissolves that oxide and chemically “wakes up” the surface. On many decorative jobs this acid dip also neutralizes any last traces of alkaline cleaner. Because activated metal re-oxidizes quickly, this is the *last* step before the nickel tank (with only a guard rinse between).

★ **REMEMBER**

No activation, no adhesion. Activated metal re-oxidizes in minutes — so activation comes immediately before nickel, and you don’t let activated parts sit.

• MAKE-UP & KEY PARAMETERS

PARAMETER	RANGE / PRACTICE	NOTES
Acid	Dilute H_2SO_4 or HCl, or proprietary acid salt	Per TDS
Concentration	Per TDS (often 5–15% v/v)	Titrate/monitor
Temperature	Ambient	—
Time	15–60 s	Just enough to de-oxidize; don’t over-etch
Agitation	Light	Reaches recesses

⚠ **HEADS-UP — STRONG ACID**

Acid burns on contact and damages eyes; some acids release irritating fumes. Goggles, face shield, gloves, apron, boots. **Add acid to water, never water to acid** when making up. Skin/eye contact → flush 15 minutes and get help.

⚙ **TECHNICAL STUFF — SOMETIMES A “STRIKE” SITS HERE**

On hard-to-plate base metals (zinc die-cast, some stainless or aluminum) a separate **strike** (a thin, fast initial deposit — e.g., a Woods nickel strike or copper strike) may follow activation to guarantee adhesion before the main nickel. Whether your line has one depends on the base metal; follow your process sheet.

• STEP-BY-STEP PROCEDURE

1. Confirm acid type, concentration, and that it’s in range.
2. Immerse the part for the specified short time — *don’t over-dip.*
3. Lift and drain over the activation tank.
4. Move straight to the guard rinse, then promptly to nickel — don’t let the part sit.

TEACH-BACK — CAN YOU ANSWER?

- What the acid dip removes and why (oxide → bondable surface).
- Why activated parts can’t sit (re-oxidation).
- The acid hazard and the acid-to-water rule.
- When a strike might be added (hard-to-plate base metals).

TOP MISTAKES TO AVOID

- Over-dipping (can etch/dull the surface).
- Under-dipping (oxide remains; adhesion suffers).
- Letting activated parts sit and re-oxidize.
- Carrying alkaline drag-in (kills the acid).
- Skipping PPE on “just a quick dip.”

SIGN-OFF — STATION 03

Trainee: ____________________ Trainer: ____________________ Dip time: ________ s

“I confirm I activated the part for the specified time in in-range acid, wore PPE, and moved it promptly toward nickel without letting it re-oxidize.”

STATION 04 OF 10

RINSE (GUARD RINSE)

"PROTECT THE NICKEL BATH — IT'S EXPENSIVE AND FUSSY."

Purpose. Rinse acid drag-out off the part so it doesn't contaminate the nickel bath.

• THE "WHY"

The nickel bath is a carefully balanced, expensive chemistry sensitive to pH and contamination. Acid dragged in upsets pH and introduces contaminants; the guard rinse protects it. As always, *lower conductivity = cleaner.*

• MAKE-UP & KEY PARAMETERS

PARAMETER	RANGE	NOTES
Temperature	Ambient	—
Time	20–45 s with agitation	Longer for recesses
Conductivity	< 500 µS/cm (target < 300)	Protects nickel bath
Flow	Counter-current / overflow	Saves water

⚠ **HEADS-UP**

Mild residual acid plus wet floors — goggles, gloves, non-slip footwear. Move promptly: a rinsed, activated part should go *straight* into nickel before it re-oxidizes.

• STEP-BY-STEP PROCEDURE

1. Confirm water flow and conductivity.
2. Immerse with agitation 20–45 s.
3. Drain over the rinse tank.
4. Transfer immediately to the nickel tank.

TEACH-BACK — CAN YOU ANSWER?

- Why protect the nickel bath from acid drag-in?
- What does conductivity tell you?
- Why move straight to nickel?

TOP MISTAKES TO AVOID

- Letting the part sit and re-oxidize after rinsing.
- Weak flow.
- Ignoring conductivity.
- Not rinsing recesses.

SIGN-OFF — STATION 04

Trainee: ____________________ Trainer: ____________________ Conductivity: ________ µS/cm

"I confirm the guard rinse was flowing and in-limit, and I transferred the part straight to nickel without letting it re-oxidize."

💡 **TIP**

This is the rinse standing between a cheap acid tank and your most expensive, fussiest chemistry — the nickel bath. Treat it as the bouncer at the nickel tank's door: nothing acidic or dirty gets past it.

STATION 05 OF 10 · THE REAL WORK

NICKEL PLATE — THE UNDERCOAT

“THE BATH GETS CALLED ‘CHROME,’ BUT THE NICKEL DOES THE WORK.”

Purpose. Build the **nickel undercoat** — bright (and often duplex/triplex) nickel that provides the part’s **thickness, brightness, leveling, and corrosion protection.**

• THE “WHY”

This is the most important plating tank on the line. The chrome flash later is thin and cosmetic; **the nickel is what makes the part look like a mirror and what protects it from corrosion for years.** Get the nickel right and the part is most of the way to a great finish. Get it wrong — dull, hazy, thin, pitted, poorly leveled, or with the wrong duplex structure — and no chrome flash can save it.

The cast at this tank:

- **Cathode** — *your part* (negative). Where nickel lands.
- **Anode** — **soluble nickel** (rounds, chips in titanium baskets, or bars). They dissolve to feed nickel back into the bath.
- **Electrolyte** — a **Watts-type nickel bath**: nickel sulfate + nickel chloride + boric acid + brighteners/levelers + wetting agent.

★ **REMEMBER**

The nickel undercoat is where the real work happens. Brightness, leveling, thickness, and corrosion protection are all set here. When this manual says “focus on the nickel,” this is the tank it means.

⚠ **HEADS-UP — NICKEL SENSITIZATION, AEROSOL, AND BORIC ACID**

The nickel tank is the “quiet” hazard. Nickel salts can cause **lifelong allergic dermatitis** and the hot tank throws a **nickel-bearing aerosol** that you shouldn’t breathe; some nickel compounds are carcinogenic by inhalation. The **boric acid** in the bath is a skin/eye irritant and a reproductive-hazard concern (REACH SVHC). **Wear gloves, keep solution off your skin, use the tank ventilation, and don’t make boric-acid additions in a way that raises dust.**

💡 **TIP — BRIGHTNESS IS BUILT, NOT SPRAYED ON**

If parts come out hazy, dull, or “milky,” your instinct might be to blame the chrome. Nine times out of ten it’s the **nickel** — low brightener, off pH, wrong temperature, or contamination. Diagnose the nickel first.

• THE BATH MAKE-UP

TYPICAL BRIGHT WATTS · VERIFY VS. TDS

COMPONENT / PARAMETER	TYPICAL RANGE	WHAT IT DOES
Nickel sulfate ($NiSO_4 \cdot 6H_2O$)	225–300 g/L	Main source of nickel metal
Nickel chloride ($NiCl_2 \cdot 6H_2O$)	40–60 g/L	Helps anodes dissolve; boosts conductivity
Boric acid (H_3BO_3)	30–45 g/L	**pH buffer** — keeps deposit bright/sound
Brighteners (Class I + II)	Per TDS	Grain refinement → mirror brightness
Leveler	Per TDS	Fills micro-scratches → smoother than input
Wetting agent / anti-pit	Per TDS	Releases hydrogen bubbles → prevents pitting
pH	~3.8–4.5	Too low = dull/etched; too high = hazy/rough
Temperature	~55–65°C (130–150°F)	Affects brightness, ductility, deposition
Cathode current density	~3–5 A/dm² (≈30–50 ASF)	Drives deposition; too high = "burn"
Cathode efficiency	~93–97%	Nearly all current makes nickel
Anodes	Soluble nickel (often Ti baskets)	Feed nickel back into the bath
Agitation / Filtration	Air or mechanical / continuous	Even deposit; removes pit-causing particles

⚙ TECHNICAL STUFF — WHY BORIC ACID (AND PH) MATTER SO MUCH

At the part surface, plating consumes hydrogen ions, so the local pH wants to *rise*. If it rises too far you get cloudy, rough, or burnt nickel and even nickel-hydroxide inclusions. **Boric acid buffers** the surface pH, holding it steady so the deposit stays bright and sound. That's why boric acid isn't optional and why bath pH is one of the most-checked numbers on the line.

⚙ TECHNICAL STUFF — DUPLEX AND TRIPLEX NICKEL (PREVIEW)

High-corrosion-spec parts (especially automotive exterior) don't use a single nickel layer. They use **duplex nickel:** a **semi-bright** (sulfur-free, ductile) layer first, then a **bright** (sulfur-containing) layer on top. The two layers have slightly different electrochemical potentials, so any corrosion that reaches the nickel **spreads sideways along the boundary instead of digging straight down to the steel. Triplex** adds a thin high-sulfur "post" nickel for even better control. This is *the* core corrosion trick of decorative chrome — full treatment in Part Three.

• STEP-BY-STEP PROCEDURE

1. **Check the bath:** temperature, pH, and (per shift) brightener/chemistry status from the log. Confirm anodes are present and baskets full, filtration and agitation running.
2. **Confirm the part** came straight from a clean guard rinse, activated and not re-oxidized.
3. **Rack/lower the part** for good electrical contact and even exposure to the anodes.
4. **Set the current** to the target current density for the part's surface area (your lead/process sheet gives the amps).
5. **Plate for the time** needed to reach target thickness (efficiency is high, so nickel builds steadily — your process sheet gives the time/amp-hours).
6. **Watch for:** even bright deposit, no burning at edges, no gas pitting. Adjust per your lead if something looks off.
7. **Withdraw slowly,** draining over the tank.
8. **Move promptly** to the post-nickel rinse and on toward chrome — **don't let the nickel passivate** (it does so fast; see Station 07).

• TOP MISTAKES

1. Blaming the chrome for haze/dullness that's really a nickel problem (pH, brightener, temp, contamination).
2. Running pH out of range (dull/etched if low; rough/hazy if high).
3. Wrong current density — too low (slow, possibly dull) or too high (burnt edges).
4. Letting the nickel-plated part sit and **passivate** before chrome (causes chrome skip/coverage defects).
5. Skipping gloves/ventilation and getting sensitized to nickel.
6. Letting filtration lapse — particles cause roughness and pits.

★ **REMEMBER**

Two clocks are always running once the nickel is plated: the **passivation clock** (nickel grows an invisible oxide in minutes that chrome won't cover) and the **quality clock** (a wet idle part can spot or stain). Both say the same thing: *keep the part moving toward chrome.*

☀ **FUN FACT**

The leveling additives in a bright-nickel bath are so good that a part can come out of the tank *smoother* than it went in — the nickel literally fills the tiny valleys faster than the peaks. That's why a lightly buffed part can emerge looking like polished glass. The mirror isn't the chrome; it's the nickel doing optical magic.

• TEACH-BACK CHECKLIST

- [ ] Explain *why the nickel undercoat — not the chrome — does the corrosion protecting.*
- [ ] Name the four key Watts components and what each does (Ni sulfate, Ni chloride, boric acid, brighteners).
- [ ] State the bath pH, temperature, and current-density ranges.
- [ ] Explain why boric acid (and stable pH) matter.
- [ ] Describe duplex nickel in one sentence and why it improves corrosion life.
- [ ] State the nickel/boric-acid hazards and the required PPE.
- [ ] Explain why a nickel-plated part can't sit before chrome (passivation).

• QUICK TROUBLESHOOTING

SYMPTOM	PROBABLE CAUSE	WHAT TO DO
Hazy / dull / milky deposit	Low brightener, off pH, wrong temp, contamination	Check nickel chemistry first — not the chrome
Rough / pitted deposit	Low wetter/anti-pit; particles; filtration lapse	Check anti-pit and restore filtration
Burnt edges	Current density too high	Lower CD; improve racking
Blistering / peeling	Poor clean/activation upstream	Re-verify Stations 01–03
Chrome skips later	Nickel passivated before chrome	Move faster; verify Station 07 activation

SIGN-OFF — STATION 05

Trainee: ____________________ Trainer: ____________________ pH: ________ Temp: ________°C

"I confirm the nickel bath was in range (pH, temperature, current density), the part came in clean and activated, I plated to the specified thickness with even brightness, and I moved it promptly toward chrome without letting it passivate. I wore nickel/boric-acid PPE and used ventilation."

TIP

Keep a mental checklist of the four numbers that decide nickel quality — **pH, temperature, current density, and brightener level.** When a part comes out wrong, walk those four before you ever touch the chrome tank. The answer is almost always in one of them.

STATION 06 OF 10

RINSE (PRE-CHROME)

"WASH THE NICKEL OFF — AND KEEP IT OUT OF THE CHROME TANK."

Purpose. Rinse the nickel-solution film off the part before it enters the chrome bath, and (often) **recover** dragged-out nickel.

THE "WHY"

Nickel dragged into the chrome bath is a contaminant — especially harmful to a **trivalent** chrome bath, which is very sensitive to metallic contamination. Even in hexavalent chrome, you don't want nickel solution diluting/contaminating the bath, and you don't want to waste nickel. This rinse protects the chrome chemistry and recovers nickel. **But** — this rinse is also where the **passivation clock** is ticking, so it must be brief and the part must move on quickly.

💡 **TIP — RECOVERY (DRAG-OUT) RINSE**

A **still or low-flow recovery rinse** placed right after nickel captures dragged-out nickel in concentrated form, which can be returned to the bath — saving nickel and reducing what goes to waste treatment.

MAKE-UP & KEY PARAMETERS

PARAMETER	RANGE	NOTES
Temperature	Ambient	—
Time	15–40 s with agitation	Keep brief — passivation clock
Conductivity	Monitor	Protects chrome bath
Flow	Recovery + counter-current	Captures nickel; saves water

⚠ **HEADS-UP**

Rinse water carries nickel — a sensitizer. Gloves, goggles, non-slip footwear. Don't dawdle: the nickel surface is passivating while you stand there.

STEP-BY-STEP PROCEDURE

1. Confirm flow/recovery setup and conductivity.
2. Rinse briefly with agitation.
3. Drain over the tank.
4. Move **straight** to chrome activation/strike — minimize the delay.

TEACH-BACK — CAN YOU ANSWER?

- Why keep nickel out of the chrome bath (especially trivalent)?
- What is a recovery rinse for?
- Why must this rinse be brief?

TOP MISTAKES TO AVOID

- Rinsing so long the nickel passivates.
- Weak/contaminated rinse letting nickel into chrome.
- Not using recovery to save nickel.
- Dawdling between nickel and chrome.

SIGN-OFF — STATION 06

Trainee: ____________________ Trainer: ____________________

"I confirm I rinsed the nickel film (using recovery where provided), kept it brief to avoid passivation, and moved the part straight to chrome activation."

STATION 07 OF 10

CHROME ACTIVATION / STRIKE

"RE-WAKE THE NICKEL — OR THE CHROME WON'T COVER."

Purpose. Re-activate the freshly-plated nickel so the chrome bonds and covers evenly, with no bare "skip-plate" spots.

• THE "WHY"

Nickel **passivates** (forms a thin invisible oxide) within minutes of leaving the nickel tank. Chrome plates poorly on passivated nickel — you get bare patches, especially in recesses (low-current-density areas), and adhesion suffers. So before chrome, the nickel is **re-activated.** This can be:

- a short **cathodic or acid activation** in a dedicated strike tank, or
- the **leading "strike" portion of the chrome cycle itself** — many hex chrome processes briefly hit the part at higher current or use the bath's initial contact to activate and start covering. Trivalent processes have their own recommended activation.

★ **REMEMBER**

The reason chrome skips bare in recesses is almost always passivated nickel and chrome's poor throwing power — not "not enough chrome." Fix it with proper activation and good racking/anode setup, not by blasting more current.

• MAKE-UP & KEY PARAMETERS

PARAMETER	RANGE / PRACTICE	NOTES
Method	`Acid/cathodic strike, or in-chrome strike step`	Per process
Acid (if separate)	`Dilute, per TDS`	De-passivates nickel
Time	`Short — seconds to ~1 min`	Just enough to activate
Transfer to chrome	`Immediate`	Don't let nickel re-passivate

⚠ **HEADS-UP**

If activation happens *in the hexavalent chrome tank,* you are already working over **carcinogenic Cr VI chromic acid** — mist suppression, ventilation, respirator (per program), and PPE apply *now,* not just at "plating." If it's a separate acid strike, strong-acid burn/fume hazards apply.

• STEP-BY-STEP PROCEDURE

1. Confirm whether activation is a separate strike or part of the chrome cycle, and set up accordingly.
2. If separate: activate for the short specified time, then transfer to chrome **immediately.**
3. If in-chrome: confirm the strike step (current/time) per the process before the main chrome cycle.
4. Verify Cr VI controls (if hex) before energizing.

TEACH-BACK — CAN YOU ANSWER?

- Why must nickel be re-activated before chrome (it passivates fast)?
- What causes "skip-plate" bare spots in recesses?
- What Cr VI controls apply if activation is in the hex chrome tank?

TOP MISTAKES TO AVOID

- Letting the nickel passivate before chrome (#1 coverage defect cause).
- Skipping/short-changing activation, then blaming the bath.
- Treating an in-chrome strike as "not really chrome yet" (hex Cr VI controls).
- Wrong strike current/time per the process.

SIGN-OFF — STATION 07

Trainee: ____________________ Trainer: ____________________ In-chrome strike? Y / N

"I confirm I re-activated the nickel for the specified time and moved straight into chrome with full Cr VI controls (if hex), without letting the nickel re-passivate."

STATION 08 OF 10 · THE BRIGHT FLASH

DECORATIVE CHROME PLATE

"THE THINNEST LAYER ON THE PART — AND, IF HEXAVALENT, THE MOST DANGEROUS TANK ON THE LINE."

Purpose. Deposit the **thin, bright chromium flash (0.13–0.5 µm)** that gives the part its cool bright tone, tarnish resistance, hardness, and — when **micro-discontinuous** — improved corrosion life.

• THE "WHY"

The chrome cap is what the customer *sees* and touches. It keeps the nickel from tarnishing, adds scratch/wear resistance, and (engineered as microcracked or microporous) spreads corrosion into a fine, shallow pattern instead of deep pits. It is **thin** because it only needs to be a bright, hard skin — the nickel underneath does the heavy corrosion work.

The cast depends on which chemistry you run:

Hexavalent (Cr VI): cathode = your part (negative); anode = **insoluble lead/lead-tin** (carries current; keeps the bath oxidized; does NOT feed chromium); electrolyte = **chromic acid (CrO_3) + catalyst (sulfate, ~100:1)**, run cooler than hard chrome (~35–45°C), orange-yellow, **carcinogenic.**

Trivalent (Cr III): cathode = your part; anode = **special inert/coated or graphite anodes** (often in a shielded/divided cell to keep the bath from oxidizing its Cr^{3+} back to Cr^{6+}); electrolyte = **trivalent chromium salts + complexers + conductivity salts + buffers (often boric)**, run cooler (often near room temp to ~45°C), **no Cr VI in the bath.**

⚠ HEADS-UP — IF THIS IS A HEXAVALENT TANK, IT IS THE SINGLE MOST DANGEROUS TANK ON THE LINE

Chromic acid that throws an inhalable, **carcinogenic Cr VI mist,** corrosive liquid, high amperage. **Mist suppression (fume suppressant/foam blanket) + local exhaust (push-pull) must be working before you run it.** Respirator per program. Full PPE. No eating/drinking/face-touching. If the foam is gone or you can see mist rising, the tank is **unsafe — stop and report.** OSHA 29 CFR 1910.1026 applies (PEL 5 µg/m³, action level 2.5 µg/m³). The thin deposit does **not** make the bath any less carcinogenic.

⚠ HEADS-UP — IF THIS IS A TRIVALENT TANK, LOWER HAZARD IS NOT NO HAZARD

No Cr VI carcinogen in the bath — but it still holds **metal salts, boric acid, and complexers** that burn/irritate skin and eyes and need waste treatment. Wear full chemical PPE, keep solution off your skin, and use ventilation. Boric acid handling cautions apply.

• BATH MAKE-UP — HEXAVALENT DECORATIVE

VERIFY VS. TDS

COMPONENT / PARAMETER	TYPICAL RANGE	WHAT IT DOES
Chromic acid (CrO_3)	~250–400 g/L	Source of all the chromium
Catalyst — sulfate (SO_4)	Set by ratio	Makes deposition possible
CrO_3 : SO_4 ratio	~100 : 1	The heart of the bath — keep in range
Mixed/fluoride catalyst (some)	Per TDS	Better coverage/efficiency
Temperature	~35–45°C (95–113°F)	Cooler than hard chrome
Cathode current density	~10–25+ A/dm²	Lower CD/time than hard chrome
Deposit thickness	0.13–0.5 µm	Thin bright flash
Plating time	Seconds to a few minutes	Very short vs. hard chrome
Cathode efficiency	~10–20%	Most current makes gas
Anodes	Insoluble lead/lead-tin	Carry current; keep bath oxidized
Mist suppressant	Fume suppressant / foam blanket	Cuts Cr VI mist at the source

• BATH MAKE-UP — TRIVALENT DECORATIVE

VARIES BY SUPPLIER

COMPONENT / PARAMETER	TYPICAL PRACTICE	WHAT IT DOES
Trivalent Cr source	Cr^{3+} sulfate or chloride	Source of chromium (already +3)
Complexers / conductivity salts	Per TDS	Keep Cr^{3+} soluble & platable
Buffer (often boric)	Per TDS	Holds pH
Temperature	~25–45°C	Often near room temperature
Current density	Per TDS (often moderate)	More forgiving coverage
Deposit thickness	0.13–0.5 µm	Thin bright flash (same goal)
Anodes	Special inert/coated/graphite	Avoid oxidizing $Cr^{3+} \rightarrow Cr^{6+}$
Contamination control	Tight	Metals (Ni, Cu, Fe, Zn) poison it

★ **REMEMBER — THE CHROME NUMBERS TO CARRY (HEX DECORATIVE)**

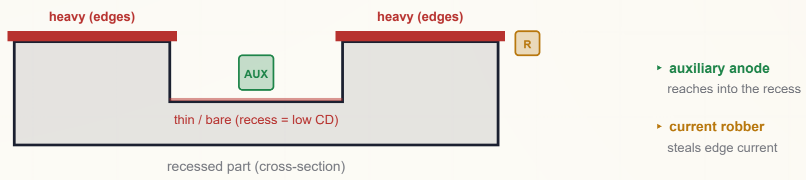
CrO_3 ~250–400 g/L, **ratio ~100:1**, **temp ~35–45°C**, **CD ~10–25+ A/dm²**, **thickness 0.13–0.5 µm**, efficiency ~10–20%, **insoluble lead anodes**, **Cr VI = carcinogen.** These let you sanity-check the tank.

⚙ TECHNICAL STUFF — WHY DECORATIVE HEX RUNS COOLER AND SHORTER THAN HARD CHROME

Same chromic-acid chemistry, different goal. Hard chrome wants a *thick, hard* build, so it runs hotter and for hours. Decorative chrome wants a *thin, bright* flash, so it runs cooler and for seconds-to-minutes. The bright “decorative” plating window (temperature × current density) is tuned for appearance, not thickness. Stay in your process window or you'll get dull, milky, smoky, or “rainbow” chrome.

THROWING POWER & COVERAGE (FULL TREATMENT IN PART THREE)

Chrome's **throwing power is poor:** it piles onto edges/high spots and starves recesses and far surfaces, giving bare “skip-plate” spots in low-current-density areas. You manage it with **good racking/orientation, auxiliary (conforming) anodes** reaching into recesses, and **current robbers/shields** to tame edges. **Trivalent's better throwing power** is one of its real advantages here.



• STEP-BY-STEP PROCEDURE

1. **Verify controls (hex):** foam blanket present, exhaust pulling, respirator/PPE on. (Trivalent: PPE/ventilation on.)
2. **Confirm bath in range:** temperature, chemistry/ratio (hex) per the log; rectifier set correctly.
3. **Confirm the part** came in freshly activated (Station 07) and not re-passivated.
4. **Energize** at the specified current; run the short chrome cycle (seconds–minutes) per the process.
5. **Watch coverage:** even bright deposit, no smoky/dull/rainbow areas, no bare recesses.
6. **De-energize, withdraw slowly,** draining over the tank.
7. **Move to drag-out/rinse** promptly.

• TOP MISTAKES

1. **Running a hex tank with dead mist suppression or weak exhaust** — a serious Cr VI exposure, not just a quality issue. Stop and report.
2. Bare/skip-plate recesses — usually passivated nickel (Station 07) or poor throwing power, *not* “more current.”
3. Dull, smoky, or rainbow chrome — out-of-window temperature/CD or chemistry (hex ratio off; triv contamination).
4. Treating decorative hex as “low risk because it’s thin” — the *bath* is the carcinogen.
5. Letting nickel contamination into a trivalent bath (poisons coverage/color).
6. Over-running the cycle (decorative chrome is a flash, not a build).

• TEACH-BACK CHECKLIST

- [ ] State the chrome thickness range and *why it’s so thin* (the nickel does the corrosion work).
- [ ] Identify whether your shop runs **hex or trivalent** and state the key hazard of each.
- [ ] (Hex) State the bath chemistry, the ~100:1 ratio, temperature, and the Cr VI controls that must be working.
- [ ] Explain chrome’s poor throwing power and three tools to manage coverage.
- [ ] Explain why bare recesses are usually a nickel-passivation/throwing-power issue, not a “more current” issue.

FUN FACT

A decorative chrome cycle can be over in under a minute, depositing a layer thinner than a wavelength of visible light is wide in some areas — yet that whisper-thin film is what makes a bumper look like a mirror and keeps the nickel from yellowing. Big visual payoff from almost no metal.

SIGN-OFF — STATION 08

Trainee: ____________________ Trainer: ____________________ Hex / Triv (circle) · Temp: ______°C

“I confirm the chrome bath was in range, the part came in freshly activated, I plated a bright even flash with no skip/dullness, and — if hexavalent — full Cr VI controls (mist suppression, exhaust, respirator, PPE) were working the entire run.”

STATION 09 OF 10

DRAG-OUT / RINSE

"RECOVER THE CHROME, CONTAIN THE HAZARD, KEEP IT OUT OF THE DRAIN."

Purpose. Wash the chrome-solution film off the part and **recover** dragged-out chrome — critical for cost and, on a hex line, for keeping a carcinogen contained.

THE "WHY"

Chrome solution clings to the part as it leaves the tank (drag-out). On a hexavalent line that drag-out is **carcinogenic Cr VI** — recovering it saves expensive chromic acid *and* keeps Cr VI out of the rinse water, the floor, and the environment. A **still/recovery rinse first** captures concentrated chrome for return or treatment; then flowing rinses clean the part.

MAKE-UP & KEY PARAMETERS

PARAMETER	RANGE	NOTES
First rinse	`Still / recovery (drag-out)`	Captures concentrated chrome
Following rinses	`Counter-current flowing`	Clean the part
Temperature	`Ambient`	—
Time	`20–60 s with agitation`	Flush recesses well

⚠ **HEADS-UP — HEX LINE: THIS WATER IS CARCINOGENIC**

On a hexavalent line the drag-out rinse holds **Cr VI.** Gloves, goggles, apron; **never let it reach a floor drain;** it must go to **Cr VI waste treatment (reduce $Cr^{6+} \rightarrow Cr^{3+}$, then precipitate)** — see Part Three. Slip hazard from wet floors as always.

STEP-BY-STEP PROCEDURE

1. Lift the part slowly from chrome, draining back into the tank to minimize drag-out.
2. Dip in the recovery/still rinse first.
3. Move through the flowing rinses with agitation; flush recesses.
4. Drain and proceed to dry/inspect.

TEACH-BACK — CAN YOU ANSWER?

- What is drag-out and why does it matter (cost + carcinogen)?
- What does a recovery rinse do?
- Where must Cr VI water never go?

TOP MISTAKES TO AVOID

- Pulling parts fast and slinging drag-out (waste + Cr VI spread).
- Skipping the recovery rinse.
- Letting chrome water reach a drain.
- Not flushing recesses.

SIGN-OFF — STATION 09

Trainee: ______________________ Trainer: ______________________

"I confirm I drained parts over the chrome tank, used the recovery rinse, rinsed thoroughly, and ensured chrome rinse water went to treatment — never to a drain."

STATION 10 OF 10

DRY / FINAL INSPECTION

“A MIRROR FINISH SHOWS EVERY WATER SPOT — DRY IT GENTLY, INSPECT IT HARD.”

Purpose. Dry the part **without water-spotting the bright finish** and verify appearance, coverage, and (where specified) thickness/adhesion before shipping.

• THE “WHY”

Decorative chrome is a *show* finish — water spots, stains, and handling marks are defects. A gentle, clean dry (warm air, sometimes a DI-water final rinse to prevent spotting) protects the appearance. Inspection is the last gate before the customer sees it.

• MAKE-UP & KEY PARAMETERS

PARAMETER	RANGE / PRACTICE	NOTES
Final rinse	`DI water (often warm)`	Prevents spotting
Dry method	`Warm forced air / blow-off`	Gentle; no scratching
Temperature	`Warm, not hot`	Avoid discoloration
Inspection	`100% visual; sampling for thickness/adhesion`	Per spec

⚠ **HEADS-UP**

Hot air and residual chemistry — mild burn and slip hazards; handle dried parts with clean gloves to avoid fingerprints and to keep any residual chemistry off skin.

• STEP-BY-STEP PROCEDURE

1. Give a clean DI final rinse if specified.
2. Dry with warm forced air / blow-off; don’t wipe a wet mirror finish (it scratches).
3. **Inspect:** full bright coverage (no skip/bare recesses), uniform color (no smoky/rainbow areas), no haze/dullness (a nickel tell), no pits/roughness, no water spots, good adhesion.
4. Route defects to rework/strip per your shop’s process; pass good parts to packing.

TEACH-BACK — CAN YOU ANSWER?

- Why a gentle, spot-free dry matters on a show finish.
- What each defect points to (haze → nickel; skip → activation/throwing power).
- How to handle finished parts to avoid marks and exposure.

TOP MISTAKES TO AVOID

- Hard-water spotting from skipping the DI final rinse.
- Wiping a wet mirror finish and scratching it.
- Passing hazy/dull (nickel) or skip-plated (chrome) parts.
- Bare-handed handling → fingerprints + skin exposure.

SIGN-OFF — STATION 10

Trainee: ______________________ Trainer: ______________________

“I confirm the part was dried without spotting, inspected for full bright coverage and appearance defects, and handled to protect the finish.”

PART THREE — ACROSS THE LINE
DEEP DIVE

THE NICKEL CORROSION SYSTEM

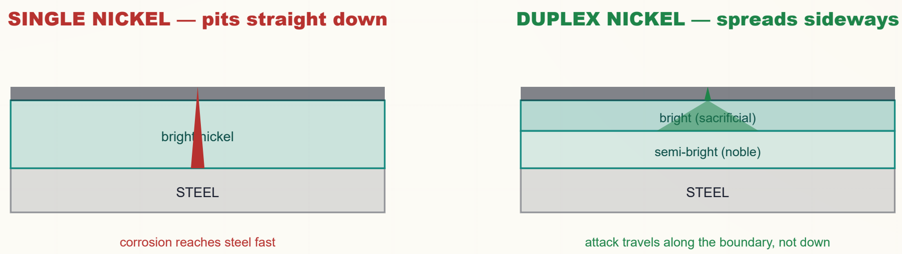
WHY A “CHROME” PART’S CORROSION LIFE IS REALLY A NICKEL STORY

The corrosion protection of a decorative chrome part comes almost entirely from the **nickel.** How long a part survives a salt-spray or CASS test (below) depends on **nickel thickness** and, crucially, **nickel structure.**

Single bright nickel protects, but high-spec parts (auto exterior especially) use **duplex** or **triplex** nickel:

- **Semi-bright nickel (bottom):** sulfur-free, ductile, more “noble” (corrosion-resistant). Plated first, directly on the base/copper.
- **Bright nickel (top):** sulfur-containing, brilliant, slightly *less* noble than the semi-bright beneath it.

Because the bright layer is slightly less noble, **it corrodes preferentially and sideways along the interface, rather than letting corrosion drive straight down to the steel.** The corrosion is forced to travel laterally through the bright nickel — a long, slow path — instead of pitting through to the base metal. This dramatically extends life. **Triplex nickel** adds a very thin **high-sulfur “post” nickel** between bright nickel and chrome to further direct and control where corrosion initiates.



REMEMBER

Duplex/triplex nickel is the core corrosion trick of decorative chrome. Two (or three) nickel layers with different sulfur contents/potentials make corrosion crawl sideways instead of digging down. The chrome on top doesn’t do this — the *nickel structure* does.

TECHNICAL STUFF — STEP TEST

Shops verify the duplex relationship with the **STEP test** (Simultaneous Thickness and Electrochemical Potential) — it measures both the thickness of each nickel layer and the potential difference between bright and semi-bright. The right potential gap (typically bright nickel ~100–150 mV more active than semi-bright) confirms the corrosion-steering effect is in place.

DEEP DIVE

MICRO-DISCONTINUOUS CHROME

TURNING THE CHROME'S POROSITY FROM A FLAW INTO A FEATURE

Plain ("crack-free" or "regular") decorative chrome has a few random pores/cracks. Where moisture finds one, it drives a **single deep pit** down through to the nickel/steel — ugly and fast-failing. The fix is to make the chrome **micro-discontinuous** — deliberately full of a *fine, uniform* pattern of either:

- **Microcracked chrome:** an engineered network of microscopic cracks — **typically a minimum of ~30 cracks per millimetre (≈300 cracks/cm) in all directions, often higher** (ASTM B456), or
- **Microporous chrome:** plated over a special nickel containing fine non-conductive particles, producing **a minimum of ~10,000 tiny pores per cm² (ASTM B456), commonly far more.**

With so many sites spread evenly, corrosion is forced to **spread out into countless tiny, shallow spots instead of a few deep pits.** The total corrosion is shared across a huge area, so the part stays cosmetically acceptable far longer and the underlying nickel is attacked only superficially.

★ **REMEMBER**

Micro-discontinuous chrome works by sharing the corrosion out. Lots of tiny, evenly-spread cracks or pores → shallow, distributed attack → much better corrosion life and appearance than a few deep pits. It's the chrome's contribution to the corrosion system — but it only works *with* a good (usually duplex) nickel underneath.

☀ **FUN FACT**

It sounds backward to *intentionally* crack or perforate your top coat for corrosion protection — but it's one of the most elegant ideas in finishing. A chrome layer with thousands of tiny, evenly-spread cracks (or pores) outlasts a "perfect" crack-free one, because it never lets corrosion concentrate in one place.

THROWING POWER & COVERAGE

Chrome's poor throwing power means **low-current-density areas (recesses, insides, far surfaces) plate thin or bare ("skip plate"),** while **edges and high spots over-build (and can burn/smoke).** Tools:

- **Racking/orientation** — position significant surfaces toward the anodes; avoid shadowing recesses.
- **Auxiliary/conforming anodes** — shaped anodes that reach *into* recesses so current (and chrome) can get there.
- **Current robbers / shields** — scrap metal or non-conductive shields that "steal" or block current at edges so they don't over-build.
- **Choose trivalent where coverage is hard** — its **better throwing power** is a real advantage on complex shapes.

💡 **TIP**

When recesses come out bare, resist the urge to crank current — that just burns the edges worse. The real fixes are activation (Station 07), racking, and auxiliary anodes.

DEEP DIVE

THE DECISION & THE WASTE

A PRACTICAL RECAP OF THE CHOICE, AND HOW TWO REGULATED HAZARDS ARE CONTAINED

• HEXAVALENT VS. TRIVALENT — THE DECISION

DRIVER	LEANS HEXAVALENT	LEANS TRIVALENT
Worker safety / carcinogen	—	**Strong advantage (no Cr VI)**
Regulatory burden (OSHA/REACH)	Heavy	**Much lighter**
Throwing power / complex parts	Moderate	**Better coverage**
Color match to "classic chrome"	**Benchmark bluish-white**	Improving; can need tuning
Color consistency / control	Mature	Needs attention
Contamination tolerance	**Tolerant**	Sensitive (metals poison it)
Color options (dark/satin)	Limited	More options available
Waste treatment	Cr VI reduction required	Simpler (no Cr VI)

★ **REMEMBER**

Trivalent is *not* automatically "better" — it's a trade-off: lower hazard and better coverage in exchange for tighter control, contamination sensitivity, and color management. The right choice depends on the parts, the specs, and the shop. *(This is a process-engineering decision; operators run whichever the shop has chosen.)*

• CR VI & NICKEL — WASTE TREATMENT, WASTEWATER & AIR

Hexavalent chrome wastewater — reduce then precipitate:

1. **Reduce** $Cr^{6+} \rightarrow Cr^{3+}$ with a reducing agent at **acidic/low pH** (e.g., sodium metabisulfite or similar — per your shop's process). This converts the carcinogenic hexavalent form to far-less-hazardous trivalent.
2. **Raise the pH** to **precipitate** the chromium as insoluble **chromium hydroxide sludge,** then **settle/filter** it out.
3. The treated water is tested before discharge per permit; the sludge is managed as hazardous waste.

Nickel wastewater: nickel is also a regulated metal — nickel-bearing rinses are typically pH-adjusted to **precipitate nickel hydroxide** and filtered; recovery rinses reduce the load. **Air:** hexavalent chrome tanks require **local exhaust + mist suppression,** often with a **fume scrubber** that washes Cr VI out of exhaust air before it leaves the building. These are required engineering controls under OSHA 1910.1026 and air permits.

⚠ **HEADS-UP**

Cr VI water must never go down a floor drain or to the sewer untreated — that's a reportable environmental release. The whole point of treatment is $Cr^{6+} \rightarrow Cr^{3+}$ then removal.

⚙ **TECHNICAL STUFF**

You can sometimes *see* the chrome reduction happening: the orange-yellow Cr^{6+} shifts toward green/blue Cr^{3+} as it reduces. Color is a clue, never a substitute for proper testing before discharge.

💡 **TIP — RINSING & WATER MANAGEMENT**

Counter-current rinsing saves large amounts of water; recovery (drag-out) rinses after nickel and after chrome capture concentrated solution for return to the bath; DI water for final rinses prevents spotting; and conductivity monitoring tells you when a rinse is doing its job (lower = cleaner). On a hex line, *every* chrome-contaminated rinse stream is a Cr VI stream and must go to treatment.

DEEP DIVE

QUALITY CHECKS

CASS, SALT SPRAY, APPEARANCE, ADHESION — HOW WE PROVE THE FINISH WILL LAST

CHECK	WHAT IT MEASURES	NOTES
Appearance	Brightness, color, coverage, defects	100% visual; haze → nickel, skip → chrome/activation
Nickel/chrome thickness	Layer thicknesses	X-ray fluorescence (XRF) or microscope cross-section; corrosion life tracks nickel thickness
Adhesion	Bond integrity	Thermal-shock (heat then quench), bend, file, or burnish tests per spec
Salt spray (ASTM B117)	Corrosion resistance	Neutral salt fog; hours-to-failure rating
CASS (ASTM B368)	*Accelerated* corrosion resistance	**C**opper-**A**ccelerated **A**cetic-acid **S**alt **S**pray — harsher/faster than B117; the benchmark for decorative nickel-chrome (auto)
STEP test	Duplex nickel potentials/thickness	Confirms the corrosion-steering structure is in place

★ REMEMBER

CASS and salt spray ultimately test the nickel system (thickness + duplex structure + micro-discontinuous chrome). When parts fail corrosion testing, look first at nickel thickness/structure, not the chrome.

☀ FUN FACT

The CASS test is so aggressive that a few days in the cabinet can simulate years of real-world exposure on a car bumper. It was developed precisely because plain salt spray was too gentle to separate a good decorative nickel-chrome system from a marginal one. If your parts pass CASS, the nickel underneath is doing its job.

💡 TIP — READING A DEFECT POINTS YOU UPSTREAM

Almost every appearance defect on a chrome part has a “home” station. **Haze/dullness** → nickel chemistry. **Skip/bare recesses** → nickel passivation or chrome throwing power. **Blister/peel** → cleaning/activation. **Water spots** → final rinse/dry. Learn the map and troubleshooting stops being guesswork.

REFERENCE

TROUBLESHOOTING QUICK-INDEX

DEFECT → CAUSE → FIX — ON THIS LINE, MOST ROADS LEAD BACK TO THE NICKEL

DEFECT	LIKELY CAUSE	FIX
Hazy / dull / milky finish	Nickel problem (low brightener, off pH, wrong temp, contamination)	Check/adjust nickel chemistry; it's almost always the nickel, not the chrome
Blistering / peeling	Poor cleaning (buffing compound/oil) or poor activation	Re-clean (water-break), verify activation; check strike for hard-to-plate metals
Bare spots / "skip plate" in recesses	Passivated nickel before chrome + poor throwing power	Re-activate nickel (Station 07); improve racking/auxiliary anodes; consider trivalent
Smoky / rainbow / off-color chrome	Out-of-window temp/CD; hex ratio off; trivalent contamination/color drift	Bring bath into process window; check ratio (hex) or contamination/color additives (triv)
Burnt / rough edges	Current density too high; throwing-power imbalance	Lower CD; add robbers/shields; rack better
Pitting	Nickel gas pitting (low wetter) or particle in bath; chrome gassing	Check nickel wetter/anti-pit & filtration; check agitation
Poor corrosion (CASS/salt-spray fail)	Thin nickel or wrong duplex structure	Increase nickel thickness; verify duplex/STEP; verify micro-discontinuous chrome
Water spots on finish	No DI final rinse / wiped wet	Add DI final rinse; warm-air dry, don't wipe
Dark/yellow tint on "chrome"	Chrome skip exposing bare nickel; or nickel tarnish	Verify chrome coverage/activation

★ **REMEMBER**

Notice how many rows say "check the nickel." On a decorative chrome line, when in doubt, **suspect the nickel tank first** — brightness, leveling, thickness, and corrosion life all live there. The chrome flash is mostly about coverage and a bright cap.

💡 **TIP**

Keep a defect-and-cause log at your station. Over a few weeks you'll see your shop's *own* recurring patterns — maybe your hazy parts always trace to a Friday-afternoon pH drift, or your skip-plate always follows a slow handoff between nickel and chrome. Your log turns this generic table into a tool tuned to your line.

⚠ **HEADS-UP**

One "defect" is never a quality problem to power through: **visible mist or dead foam on a hexavalent tank.** That's not a cosmetic issue — it's a carcinogen-exposure event. Stop, report, and don't run the tank until controls are restored.

REFERENCE

GLOSSARY — PLAIN AND SIMPLE

EVERY TERM, DEFINED THE WAY YOU'D EXPLAIN IT TO A FRIEND ON DAY ONE

- **Anode** — the positive electrode. Nickel tank: *soluble nickel,* feeds metal. Hex chrome tank: *insoluble lead,* carries current only.
- **Boric acid** — pH buffer in the nickel bath; also a handling hazard (REACH SVHC).
- **Brightener** — organic additive that makes nickel reflective.
- **CASS test** — Copper-Accelerated Acetic-acid Salt Spray (ASTM B368); accelerated corrosion test, the auto-trim benchmark.
- **Cathode** — the negative electrode; *your part,* where metal lands.
- **Cathode efficiency** — % of current that becomes metal. Nickel ~93–97%; chrome ~10–20%.
- **Chromic acid** — CrO_3 in water; the hexavalent chrome bath; **carcinogenic.**
- **Cr VI / Cr III** — hexavalent (carcinogen) vs. trivalent (far less hazardous) chromium.
- **Decorative chrome** — thin bright chrome (0.13–0.5 µm) over nickel, for appearance + corrosion.
- **Drag-out / drag-in** — solution clinging to a part leaving a tank / contaminating the next tank.
- **Duplex nickel** — semi-bright + bright nickel layers that steer corrosion sideways.
- **Leveler** — additive that fills micro-scratches so the deposit is smoother than the input.
- **Micro-discontinuous chrome** — microcracked or microporous chrome that spreads corrosion into many shallow sites.
- **PEL / Action Level (Cr VI)** — 5 µg/m³ / 2.5 µg/m³ (OSHA 1910.1026).
- **Passivation (of nickel)** — fast invisible oxide on fresh nickel that must be re-activated before chrome.
- **Salt spray (B117)** — neutral salt-fog corrosion test.
- **Sensitization** — developing a lifelong allergy (nickel dermatitis).
- **STEP test** — confirms duplex nickel potentials/thickness.
- **Strike** — a thin fast initial deposit to ensure adhesion (e.g., on zinc die-cast).
- **Throwing power** — ability to plate evenly into recesses. Nickel good; chrome poor.
- **Triplex nickel** — duplex plus a thin high-sulfur "post" nickel.
- **Watts nickel** — the classic bright-nickel bath (Ni sulfate + Ni chloride + boric acid).

★ **REMEMBER**

If you can define **cathode, anode, electrolyte, throwing power, passivation,** and **Cr VI vs. Cr III** in your own words, you can hold a real conversation with your lead, your chemist, and your chemistry supplier. That vocabulary is half of becoming a confident operator.

TEMPLATE — PRINT ONE PER OPERATOR

OPERATOR TRAINING MATRIX

COMPETENCY TRACKED PER STATION — THE TRAINER MARKS THE HIGHEST LEVEL REACHED AND DATES IT

★ **REMEMBER**

An operator is “line-qualified” only when every station they run is signed. Nobody touches the **chrome tank**, the **nickel tank**, or the **rectifier** solo until the **safety** rows are signed. Safety qualification comes first, always.

Legend: O = Observed (watched the task done correctly) · **A** = Assisted (did it with help) · **Q** = Qualified (did it solo, correctly, signed off) · **T** = Trainer (can teach/sign off others).

STATION / COMPETENCY	O	A	Q	T	DATE	TRAINER
01 Clean / Degrease (water-break)	☐	☐	☐	☐	______	______
02 Rinse (firewall / conductivity)	☐	☐	☐	☐	______	______
03 Acid Activation	☐	☐	☐	☐	______	______
04 Guard Rinse	☐	☐	☐	☐	______	______
05 Nickel Plate (the undercoat)	☐	☐	☐	☐	______	______
06 Pre-Chrome Rinse / recovery	☐	☐	☐	☐	______	______
07 Chrome Activation / Strike	☐	☐	☐	☐	______	______
08 Chrome Plate (hex/triv)	☐	☐	☐	☐	______	______
09 Drag-Out / Rinse	☐	☐	☐	☐	______	______
10 Dry / Inspect	☐	☐	☐	☐	______	______
Cr VI safety (if hex line)	☐	☐	☐	☐	______	______
Nickel/boric-acid safety	☐	☐	☐	☐	______	______
Emergency / LOTO / spill response	☐	☐	☐	☐	______	______
Waste treatment awareness	☐	☐	☐	☐	______	______

Operator name: ______________________ **Hire/start date:** ______________

Supervisor: ______________________ **Review cycle:** ☐ Annual ☐ Other: __________

⚠ **HEADS-UP**

Safety competencies (Cr VI if applicable, nickel/boric handling, emergency response) must reach **Q** before independent work — they are gates, not gradual goals. An operator may not work *any* station until those rows are signed.

PART FOUR — CERTIFICATION
20 QUESTIONS · PASSING SCORE 80% (16/20)

DECORATIVE CHROME COMPLETION TEST

COVERS EVERY STATION PLUS SAFETY — ANSWER IN YOUR OWN WORDS WHERE ASKED

★ **REMEMBER**

Passing score: 80% (16 of 20 correct). Any **safety** question answered wrong (**Q3, Q7, Q12, Q16, Q19**) is an automatic fail of the safety gate — re-teach and re-test before sign-off, regardless of total score. The Answer Key follows — no peeking until you're done!

Name: ______________________ **Date:** ____________ **Score:** _______ / 20 **Safety-gate Qs (all must be correct): 3, 7, 12, 16, 19**

Q1. On a decorative chrome part, the layer that does most of the *corrosion protecting* is:
A. The chromium top layer B. The nickel undercoat C. The base metal D. The clear lacquer

Q2. A typical decorative chrome *thickness* is about:
A. 50–100 µm B. 10–20 µm C. 0.13–0.5 µm D. 2–5 mm

Q3. (Short answer) [SAFETY GATE] Decorative chrome can be plated from two chemistries. Name both, and state which one is a **confirmed human carcinogen** and the OSHA standard/PEL that governs it.

Q4. In the *nickel* tank, the anodes are:
A. Insoluble lead B. There are no anodes C. Graphite only D. Soluble nickel that feeds the bath

Q5. The classic bright-nickel ("Watts") bath contains nickel sulfate, nickel chloride, and:
A. Cyanide B. Chromic acid C. Sodium hydroxide D. Boric acid (a pH buffer)

Q6. Typical bright-nickel bath conditions are about:
A. pH ~3.8–4.5, ~55–65°C, ~3–5 A/dm² B. pH ~13, room temp, 0.5 A/dm² C. pH ~1, 90°C, 60 A/dm² D. pH ~7, freezing, 100 A/dm²

Q7. (Short answer) [SAFETY GATE] Name **two** hazards of the *nickel* tank (the "quiet" tank) and the controls that protect you.

Q8. Why is the chromium layer so thin on a decorative part?
A. Chrome is too expensive to plate thick B. The nickel underneath does the corrosion work; chrome is a thin bright cap C. Thick chrome is illegal D. The bath can't deposit more than 0.5 µm ever

Q9. "Duplex nickel" (semi-bright then bright) improves corrosion life because:
A. It's thicker only B. The two layers' potential difference makes corrosion spread sideways instead of driving down to the steel C. It adds copper D. It removes the need for chrome

Q10. "Micro-discontinuous" (microcracked/microporous) chrome improves corrosion resistance by:
A. Sealing the surface perfectly B. Adding thickness C. Making the part harder D. Spreading corrosion into many tiny shallow sites instead of a few deep pits

Q11. Nickel-plated parts must be **re-activated** before chrome because:
A. Nickel passivates quickly and chrome won't bond/cover on passivated nickel B. The nickel is too hot C. Chrome dissolves nickel D. It isn't necessary

Q12. (Short answer) [SAFETY GATE] If your shop runs **hexavalent** decorative chrome, list the engineering control that comes **before** PPE for the mist hazard, and state where Cr VI rinse water must **never** go.

Q13. Hexavalent decorative chrome's cathode efficiency is roughly:
A. 95–98%   B. 60–80%   C. 10–20%   D. 100%

Q14. A part keeps coming out **bare in deep recesses** ("skip plate"). The best fix is usually:
A. Improve nickel activation, racking, and auxiliary anodes (chrome's throwing power is poor)   B. Crank the current way up   C. Add cyanide   D. Heat the rinse

Q15. Compared with hexavalent, **trivalent** decorative chrome generally offers:
A. Lower hazard and better throwing power, but more contamination sensitivity and color-control work   B. Higher carcinogen risk   C. Thicker deposits   D. No anodes

Q16. (Short answer) [SAFETY GATE] Describe the **reduce-then-precipitate** method for treating hexavalent-chrome wastewater (what each step does and the goal).

Q17. The standard accelerated corrosion test for decorative nickel-chrome (especially auto trim) is:
A. Tensile test   B. CASS (Copper-Accelerated Acetic-acid Salt Spray, ASTM B368)   C. Hardness test   D. Water-break test

Q18. A part comes out **hazy/dull.** The first place to look is:
A. The chrome tank   B. The dryer   C. The rinse conductivity   D. The **nickel** chemistry (brightener, pH, temp, contamination)

> ⚠ **SAFETY SECTION — MUST BE CORRECT TO CERTIFY**
> Questions 3, 7, 12, 16, and 19 cover hazards that can permanently harm you or others. A miss on any safety item means re-teach and re-test before sign-off.

Q19. (Short answer / Safety) [SAFETY GATE] List **four** pieces of PPE for work at the chrome tank, and name the procedure required before any maintenance on a high-amperage **rectifier.**

Q20. (Short answer) Name **two** ways decorative chrome differs from *hard* (functional) chrome.

End of test. Hand to your trainer for grading.

FOR TRAINERS & SELF-CHECKERS

ANSWER KEY

SHORT-ANSWER RESPONSES DON'T NEED TO MATCH WORD-FOR-WORD — LOOK FOR THE KEY CONCEPTS IN BOLD

⚠ **HEADS-UP**

Keep this page out of the trainee's copy. Questions **3, 7, 12, 16, 19** are safety-critical — any wrong answer there requires re-training before sign-off, regardless of total score.

Q1. B — The **nickel undercoat** does most of the corrosion protecting.

Q2. C — **0.13–0.5 µm** (a thin flash).

Q3. Hexavalent (Cr VI) and **trivalent (Cr III)**. **Hexavalent is a confirmed human carcinogen,** governed by **OSHA 29 CFR 1910.1026** with a **PEL of 5 µg/m³** (action level 2.5 µg/m³).

Q4. D — **Soluble nickel** anodes that dissolve to feed the bath.

Q5. D — **Boric acid** (the pH buffer).

Q6. A — pH ~3.8–4.5, ~55–65°C, ~3–5 A/dm².

Q7. Any two: **nickel sensitization/allergic dermatitis**, **nickel aerosol** (don't breathe; some Ni compounds carcinogenic by inhalation), **boric acid** (irritant + reproductive-hazard concern). Controls: **gloves, keep solution off skin, tank ventilation, hygiene, dust control on additions.**

Q8. B — The nickel does the corrosion work; chrome is a thin bright cap.

Q9. B — The bright/semi-bright potential difference steers corrosion **sideways** instead of down to the steel.

Q10. D — Spreads corrosion into many tiny **shallow** sites instead of a few deep pits.

Q11. A — **Nickel passivates fast;** chrome won't bond/cover on passivated nickel.

Q12. Engineering controls before PPE: mist suppression (fume suppressant/foam blanket) + local exhaust/ventilation (a respirator is the last line, not the first). Cr VI rinse water must **never go down a floor drain / to the sewer untreated** — it goes to Cr VI waste treatment.

Q13. C — ~10–20% (low; most current makes gas).

Q14. A — Improve nickel activation, racking, and **auxiliary/conforming anodes** (chrome's throwing power is poor); more current just burns the edges.

Q15. A — Lower hazard + better throwing power, but more contamination sensitivity and color-control work.

Q16. Reduce $Cr^{6+} \rightarrow Cr^{3+}$ with a reducing agent at **acidic/low pH**, then **raise pH to precipitate** chromium as insoluble hydroxide sludge and **settle/filter** it out. Goal: convert the carcinogenic hexavalent form to the far-less-hazardous trivalent form and remove it before discharge.

Q17. B — **CASS** (ASTM B368).

Q18. D — The **nickel** chemistry (haze/dullness is almost always a nickel problem).

Q19. PPE (any four): sealed chemical splash goggles, face shield, chemical-resistant gloves, chemical apron/suit, chemical-resistant boots, respirator (per the Cr VI program on a hex line). Rectifier procedure: **LOTO (Lockout/Tagout).**

Q20. Any two: decorative chrome is **thin (0.13–0.5 µm) over nickel** for appearance, while hard chrome is **thick, directly on steel, functional, and ground to size**; decorative relies on a **nickel undercoat** while hard chrome has **none**; decorative hex runs **cooler (~35–45°C)** than hard chrome (~50–60°C); decorative is for **looks + tarnish/corrosion,** hard chrome for **wear/dimension.**

Answer distribution (MC): A=4 · B=3 · C=3 · D=4 *(self-check — the 14 multiple-choice keys spread evenly; Q3, Q7, Q12, Q16, Q19, Q20 are short-answer)*

★ **REMEMBER**

A score of 16/20 passes — but every safety question (3, 7, 12, 16, 19) must be correct to certify. Miss a safety item → re-teach and re-test before sign-off. On a Cr VI line, we don't gamble with people.

PP

PLATING POSTERS

PLATING POSTERS INC · METAL FINISHING REFERENCE SERIES

CERTIFICATE OF COMPLETION

DECORATIVE CHROME PLATING TRAINING PROGRAM

This certifies that

OPERATOR NAME

has successfully completed the **Decorative Chrome (Nickel-Chromium System) Plating** training course — covering the full decorative-chrome workflow (Clean, Rinse, Acid Activation, Rinse, **Nickel Plate undercoat**, Rinse, Chrome Activation/Strike, **Decorative Chrome Plate**, Drag-out/Rinse, and Dry/Inspection), the bright/duplex/triplex nickel undercoat and its central role in corrosion protection, Watts nickel chemistry, micro-discontinuous (microcracked/microporous) chrome, throwing power and coverage, the differences between **hexavalent and trivalent** chrome, CASS/salt-spray quality testing, and — above all — the safety practices required of every operator: **nickel sensitization and boric-acid handling**, and, on a hexavalent line, the **hexavalent chromium (Cr VI) carcinogen controls** (mist suppression, ventilation, respiratory protection, hygiene, spill response, and $Cr^{6+} \rightarrow Cr^{3+}$ waste treatment) — and has passed the Completion Test with a score of ________ / 20, including all required safety competencies.

This operator is hereby cleared for **independent work** at the certified stations listed on the Operator Training Matrix.

Date of completion: ____________________ **Re-certification due:** ____________________

TRAINEE SIGNATURE

CERTIFYING TRAINER

SUPERVISOR

Training reference only. Verify all parameters against your chemistry supplier's TDS, customer specifications, and applicable regulatory requirements before production use. If your shop runs hexavalent chrome, **hexavalent chromium is a confirmed human carcinogen** — always consult current SDS and comply with OSHA 29 CFR 1910.1026, EPA, REACH, and local regulations.